Project Proposal

A Data Mining Solution for Early Gallstone Prediction
to Support Sustainable Healthcare

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1 Introduction

Sustainable Development Goal 3 (SDG 3), *Good Health and Well-Being* 1, aims to improve health outcomes through prevention, early intervention, and access to effective healthcare services. Achieving this goal increasingly depends on the ability of healthcare systems to use available data efficiently and support evidence-based clinical decisions. Data mining and machine learning can contribute to this objective by identifying hidden patterns in routinely collected health information and supporting earlier detection of disease risk.



Figure 1: 17 Global Challenge for Government Transparency: The Sustainable Development Goals (SDG) 2030 Agenda

Gallstone disease is a common condition that may remain asymptomatic but can lead to complications such as acute cholecystitis, biliary obstruction, cholangitis, and pancreatitis (Fujita et al., 2023; Lammert et al., 2016). Diagnosis generally relies on clinical assessment and abdominal ultrasonography. However, identifying high-risk individuals before complications occur may support earlier investigation and improve the efficient use of healthcare resources.

This project therefore develops a data-mining solution for the early prediction of gallstone disease using demographic, comorbidity, laboratory, and body-composition information. The Gallstone dataset contains records from 319 participants, including both patients with gallstones and control participants (I. Esen et al., 2025; İ. Esen et al., 2024). Machine-learning models will be used to identify patterns associated with gallstone status and estimate individual disease risk.

The proposed solution is intended as a clinical decision-support tool rather than a replacement for medical diagnosis. By evaluating predictive performance, sensitivity, interpretability, and important risk factors, the project aims to demonstrate how data-driven healthcare can support preventive intervention, reduce avoidable complications, and improve resource allocation. In this way, the project contributes to a more proactive, efficient, and sustainable healthcare system aligned with the broader objectives of SDG 3.

2 Business and Situation Understanding

2.1 Clinical background and motivation

Gallstone disease (cholelithiasis) is a common hepatobiliary disorder in which solid deposits form within the gallbladder or biliary tract. Although many gallstones remain clinically silent, obstruction of the cystic or common bile duct can lead to biliary colic and complications such as acute cholecystitis, obstructive jaundice,

cholangitis, and gallstone pancreatitis Fujita et al., 2023; Lammert et al., 2016. A recent systematic review and meta-analysis estimated that approximately 6% of the global population has gallstones, with prevalence varying substantially by sex and geographical region Wang et al., 2024. The scale of the condition and the possibility of clinically important complications make the identification of individuals at increased risk a relevant preventive and diagnostic-support problem.

Gallstone formation is multifactorial rather than the result of a single abnormal measurement. Established or commonly reported associations include age, sex, obesity and body-fat distribution, metabolic abnormalities, dyslipidaemia, and diabetes Lammert et al., 2016. These relationships are relevant to the present dataset because its predictors combine demographic information, comorbidities, body-composition measurements, and routine laboratory markers. This creates an opportunity to investigate whether the joint pattern of routinely obtainable variables contains sufficient information to distinguish patients with and without gallstones.

In current clinical practice, abdominal ultrasonography remains a principal first-line imaging method for suspected gallstone disease, together with appropriate laboratory assessment European Association for the Study of the Liver (EASL), 2016; Fujita et al., 2023; Gutt et al., 2020; National Institute for Health and Care Excellence, 2014. Therefore, the purpose of the proposed machine learning system is *not* to replace ultrasound or provide an autonomous diagnosis. Instead, the project treats prediction as a risk-stratification and decision-support problem: a model could estimate the probability of gallstone disease from non-imaging clinical information and thereby identify individuals who may warrant closer clinical assessment or confirmatory imaging.

This direction is supported by recent work applying machine learning to clinical and metabolic information. Salem et al. combined metabolomic and clinical predictors and demonstrated that nonlinear models can capture patterns associated with gallstone status Salem et al., 2023. More directly, Esen et al. introduced the dataset used in this project and reported that models based on bioimpedance and laboratory variables could distinguish gallstone cases from controls, with their strongest models reaching approximately 85% accuracy and an area under the receiver-operating-characteristic curve (ROC–AUC) of about 0.85 İ. Esen et al., 2024. These results provide a useful benchmark, but they do not remove the need for careful validation, because the available sample is small and originates from a single clinical setting.

2.2 Business and situation objectives

The practical objective of this project is to develop and evaluate an interpretable early-risk prediction approach for gallstone disease using information that may already be available from routine clinical assessment, blood testing, and bioimpedance measurement. In a realistic healthcare workflow, the intended output would be a *risk estimate* rather than a definitive diagnostic label. The model should help prioritise patients for further assessment while maintaining sufficient transparency for its predictions to be examined by a clinician.

Accordingly, the business/situation objectives are:

1. Determine whether demographic, comorbidity, laboratory, and body-composition variables can provide useful early discrimination between patients with and without gallstones;
2. Minimise clinically undesirable false-negative predictions, because a model that frequently classifies gallstone cases as low risk would have limited screening value;
3. Identify the variables that contribute most strongly to predictions so that the resulting model remains interpretable and clinically reviewable; and
4. Produce a reproducible proof-of-concept that can justify, but not substitute for, subsequent external clinical validation.

For this iteration, business success is defined as evidence that the dataset can support a stable and interpretable risk-stratification model rather than evidence that the model is ready for clinical deployment. The analytical

success criteria are therefore set as: (i) ROC–AUC of at least 0.80 under a validation procedure that keeps training and evaluation data separate; (ii) sensitivity/recall of at least 0.80 for the gallstone class at the selected decision threshold; (iii) performance superior to simple baselines such as majority-class prediction and standard logistic regression where possible; and (iv) reporting of calibration, confusion-matrix outcomes, and important predictors in addition to overall accuracy. These thresholds are project benchmarks chosen to make the data-mining goal measurable; they are not clinical guideline thresholds.

2.3 Situation assessment

2.3.1 Available resources and data

The UCI Gallstone dataset originates from a prospective descriptive study conducted at the Internal Medicine Outpatient Clinic of Ankara VM Medical Park Hospital between June 2022 and June 2023 I. Esen et al., 2025; İ. Esen et al., 2024. The source study initially considered 454 participants and excluded individuals who had previously undergone gallbladder surgery, leaving 319 participants for analysis: 161 with gallstones and 158 controls İ. Esen et al., 2024. The near-balanced outcome distribution is advantageous for a binary classification exercise because the model is not forced to learn from a severely under-represented positive class.

The dataset contains demographic variables, comorbidities, anthropometric measurements, bioimpedance measurements, hepatic-fat information, and routine laboratory results. Examples include age, sex, body mass index (BMI), diabetes, hyperlipidaemia, total body water, extracellular and intracellular water, body-fat and lean-mass measures, visceral fat measures, glucose, lipid markers, liver enzymes, creatinine, glomerular filtration rate, C-reactive protein (CRP), haemoglobin, and vitamin D I. Esen et al., 2025; İ. Esen et al., 2024. The UCI repository describes the released data as having no missing values I. Esen et al., 2025; however, this must still be confirmed programmatically before modelling rather than assumed from documentation alone.

The dataset is computationally small and can be analysed on ordinary hardware. All development in this iteration will use Python, with reproducible scripts and notebooks used to record data validation, preprocessing, feature engineering, model fitting, and evaluation. The principal libraries are pandas and NumPy for data handling, scikit-learn for preprocessing, validation, and baseline models, and XGBoost and LightGBM for boosted-tree experiments.

2.3.2 Requirements and assumptions

Several requirements follow from the medical context. First, all predictor transformations must be fitted using training data only so that information from the evaluation set cannot leak into model development. Second, model performance must be reported with class-sensitive metrics—particularly sensitivity, specificity, precision, F1 score, ROC–AUC, and the confusion matrix—rather than accuracy alone. Third, model interpretation is required because a high-performing but opaque model would be difficult to justify as clinical decision support. Finally, the analysis should distinguish prediction from causal inference: an important feature may improve classification without being a causal determinant of gallstone formation.

A critical data-validation requirement is to verify the target encoding directly from the supplied dataset before analysis. The published study describes one numerical coding for gallstone presence, whereas the UCI repository documentation presents the class labels differently I. Esen et al., 2025; İ. Esen et al., 2024. Similarly, repository metadata and the study description should be reconciled for the final number of observations. These discrepancies do not prevent modelling, but failure to resolve them could invert the target labels or create incorrect performance interpretation.

2.3.3 Constraints, risks, and contingencies

The principal constraint is sample size. With only approximately 319 participants and a relatively large number of candidate predictors, flexible models can overfit the training data. This risk will be reduced through stratified resampling, cross-validation, regularisation or restricted model complexity, and comparison with simpler baselines. Hyperparameter tuning and feature selection must occur inside the training/resampling process rather than on the full dataset.

A second constraint is external validity. The records come from a single hospital population in Ankara, and the observed relationships may not transfer directly to other countries, ethnic groups, age structures, or healthcare settings. Consequently, even strong internal validation results should be described as proof-of-concept evidence. External validation on an independent cohort would be required before clinical use.

A third risk concerns measurement availability. Some variables, particularly detailed bioimpedance measurements, may not be collected in every routine clinical setting. To address this, the project should compare a full-feature model with a reduced model based on more routinely available demographic and laboratory variables. If predictive performance remains similar, the reduced model may provide greater practical value.

Other risks include correlated body-composition variables, unstable feature importance, hidden coding errors, and optimistic results caused by repeated experimentation on the same test data. The contingency plan is to preserve a clear evaluation protocol, report uncertainty across resamples, investigate highly correlated variables, and avoid claiming that a small improvement in a single metric represents clinically meaningful superiority.

2.4 Data-mining goals

The data-mining task is a supervised binary-classification problem in which gallstone status is the target variable. The immediate technical goals are to:

1. Audit variable types, ranges, coding, duplicates, and missing values;
2. Explore relationships between gallstone status and demographic, metabolic, laboratory, and body-composition predictors;
3. Establish interpretable baselines using logistic regression and a simple tree-based model;
4. Compare a limited set of nonlinear classifiers such as random forest and gradient boosting using consistent resampling;
5. Tune decision thresholds with particular attention to sensitivity and false-negative errors;
6. Assess probability calibration in addition to discrimination;
7. Evaluate model behaviour across clinically relevant subgroups, especially sex and age groups where sample size permits; and
8. Use interpretable feature-attribution or model-specific importance methods to identify variables that consistently contribute to predictions.

These goals align directly with the business objective: the model is useful only if it can identify higher-risk individuals with acceptable sensitivity, provide a meaningful probability estimate, and explain the factors contributing to that estimate. The strongest model will therefore not automatically be the model with the highest raw accuracy; model selection will consider discrimination, sensitivity, calibration, stability, and interpretability together.

2.5 Project plan for the current iteration

Table 1 summarises the iteration running from 21 August to 21 September 2026. Development is organised around a common Python evaluation framework so that the baseline, data-centric, and model-centric strategies can be compared fairly using the same data partitions and metrics.

Table 1: Iteration plan for Python development and evaluation, 21 August–21 September 2026.

Dates	**Main activities**	**Expected output**
21–24 August	Confirm the business objective and success criteria; audit target coding, variable types, missing values, duplicates, ranges, and provenance in Python.	Validated problem statement, data dictionary, and data-quality log.
25–28 August	Complete exploratory analysis and implement a leakage-safe Python pipeline with fixed stratified 70/15/15 partitions and repeated seeds; establish logistic-regression and tree baselines.	Reproducible baseline workflow, saved split membership, and benchmark metrics.
29 August–5 September	Apply the *data-centric strategy*: correct data-quality issues, assess transformations and outliers, engineer clinically meaningful features, compare full and reduced feature sets, and perform feature selection within cross-validation.	Audited feature sets and evidence of whether data improvements increase validation performance and stability.
29 August–8 September	In parallel, apply the *model-centric strategy*: compare logistic regression, random forest, Extra Trees, support-vector machine, XGBoost, LightGBM, and stacking; tune promising models without accessing the test set.	Tuned candidate models and a cross-validated comparison table.
9–13 September	Evaluate alternative validation strategies using repeated stratified holdout and leave-one-out cross-validation (LOOCV); assess threshold choice, calibration, sensitivity, specificity, ROC–AUC, F1, and log loss.	Robustness analysis showing the effect of evaluation strategy and decision threshold.
14–17 September	Conduct error analysis, subgroup checks, feature-importance analysis, and comparison of data-centric, model-centric, and combined strategies.	Error profiles, interpretation outputs, and a defensible preferred approach.
18–21 September	Re-run the selected workflow, verify reproducibility, consolidate results and limitations, and finalise the report and recommendations.	Reproducible Python artefacts and final iteration report.

2.5.1 Development and data-mining plan

The current iteration is scheduled from 21 August to 21 September 2026. All development will be completed in Python. The work begins with a reproducible baseline and then separates improvement activities into two complementary strategies. The data-centric strategy improves the quality and representation of the input data through validation, transformation, feature engineering, feature selection, and reduced-feature experiments. The model-centric strategy holds the prepared data and evaluation rules constant while comparing model families, hyperparameters, ensembles, calibration, and classification thresholds. A final combined strategy uses only improvements supported by validation evidence.

Figure 2 presents the project timeline. The same saved data partitions and metrics will be used across strategies to ensure that any observed improvement is attributable to the tested change rather than to a different evaluation setup.

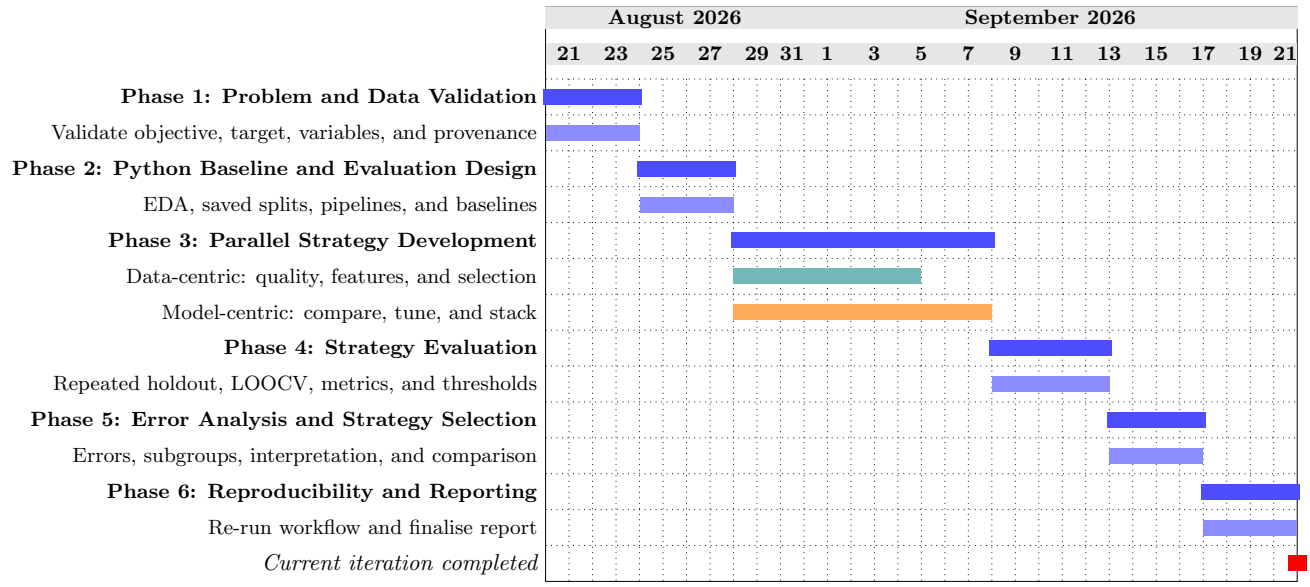


Figure 2: Python development and evaluation plan for the current gallstone prediction iteration, 21 August–21 September 2026.

The baseline strategy provides the reference result against which subsequent changes are assessed. In the data-centric stage, models are deliberately kept stable while changes to data quality, transformations, engineered variables, and feature subsets are evaluated. In the model-centric stage, the selected data representation is held stable while algorithms and hyperparameters are varied. Preprocessing and feature selection remain inside each resampling fold to prevent leakage. Repeated holdout evaluation measures variation across partitions, while LOOCV provides a complementary assessment that uses nearly all observations for training in each fold. The preferred approach will be selected using sensitivity, specificity, ROC–AUC, F1, calibration, stability, interpretability, and implementation complexity rather than accuracy alone.

Figure 3 clarifies the experimental pipeline. The untouched test partition is used only after the data-centric and model-centric choices have been made from training and validation evidence.

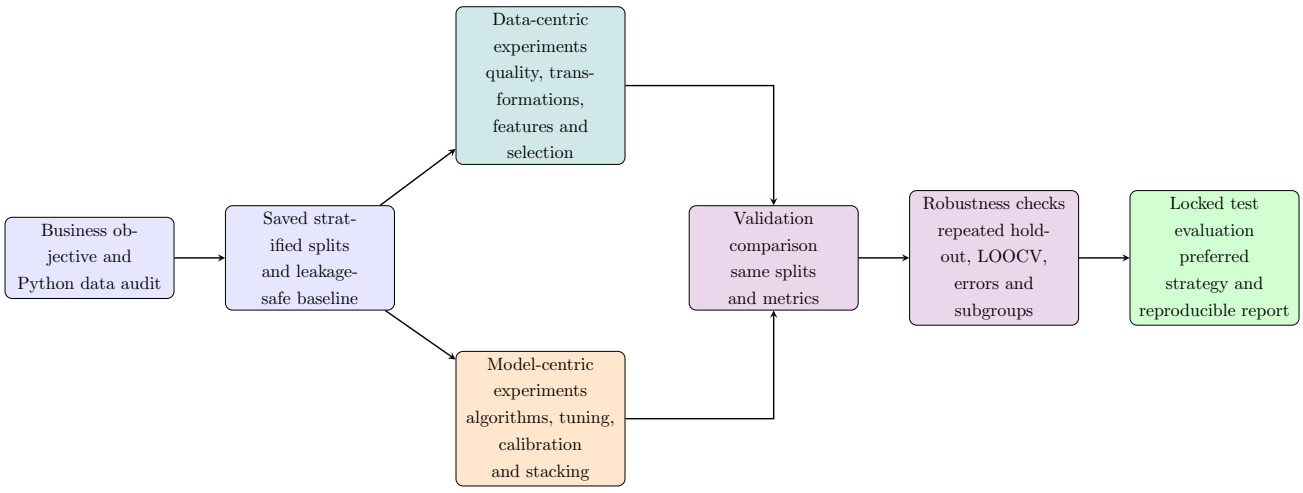


Figure 3: Python project pipeline for comparing data-centric, model-centric, and combined gallstone-prediction strategies.

2.5.2 Report writing plan

Report writing will be carried out progressively throughout the current iteration rather than being postponed until the modelling work is complete. Early sections describing the problem, dataset, and methodology will

be drafted while the corresponding analytical tasks are being completed. Results, discussion, limitations, and recommendations will then be added as model outputs become available. This approach allows the report to be updated continuously and reduces the risk of documentation being disconnected from the implemented analysis.

Figure 4 presents the report-writing schedule for 21 August–21 September 2026.



Figure 4: Day-to-day report-writing plan for the gallstone prediction project, 21 August–21 September 2026.

The completion criterion for this iteration is therefore a transparent comparison of feasible predictive models and a defensible conclusion about whether this dataset supports early gallstone-risk stratification. Clinical deployment, treatment decisions, and replacement of diagnostic imaging are explicitly outside the scope of the present iteration.

3 Data understanding

3.1 Initial Data Collection

The dataset used in this project was obtained from the UCI Machine Learning Repository and originates from the clinical study conducted by İ. Esen et al. (2024). The original data were collected prospectively from patients attending the Internal Medicine Outpatient Clinic of Ankara VM Medical Park Hospital, Turkey, between June 2022 and June 2023 (I. Esen et al., 2025; İ. Esen et al., 2024).

The original study initially collected information from 454 participants. Records from individuals who had

previously undergone gallbladder surgery were excluded, resulting in a final published dataset of 319 participants. Among these, 161 participants were diagnosed with gallstone disease and 158 formed the control group (İ. Esen et al., 2024). Data collection included demographic and clinical characteristics, comorbidities, laboratory test results, and body-composition measurements obtained through bioimpedance analysis. The resulting dataset contains 38 predictive attributes relating to factors such as age, BMI, diabetes, lipid profiles, liver-function markers, glucose levels, body water, muscle mass, and body-fat distribution/

No additional primary data were collected for the current project; therefore, this analysis represents secondary use of the published clinical dataset. The UCI repository reports no missing values. However, an issue identified during initial data collection and validation is an inconsistency in the published metadata: the repository description reports 319 participants, while one metadata field lists 320 instances. Consequently, the downloaded dataset's row count and target coding will be independently verified before further analysis.

The dataset was downloaded in tabular form and will be retained in its original form as a raw-data copy before any cleaning or transformation is performed.

3.2 Data Description

The dataset contains one binary target variable, *Gallstone Status*, and 38 predictor variables covering demographic characteristics, comorbidities, anthropometric measurements, body-composition indicators, and laboratory measurements. The predictors include both categorical and numerical variables, allowing the project to investigate multiple clinical and physiological factors that may be associated with gallstone disease.

According to the dataset documentation, no missing values are reported. However, the variable types, class coding, ranges, and missing-value patterns will be independently verified during the data-quality assessment before modelling. In particular, categorical variables stored numerically will be converted to appropriate factor or categorical representations to prevent their codes from being interpreted as continuous quantities.

Table 2 summarises the variables contained in the dataset.

Table 2: Data dictionary for the Gallstone dataset.

Variable	**Role**	**Type**	**Demo.**	**Description**	**Units**	**Missing**
Gallstone Status	Target	Binary	–	Presence or absence of gallstones	–	No
Age	Feature	Integer	Age	Age of the participant	years	No
Gender	Feature	Categorical	Gender	Gender of the participant	–	No
Comorbidity	Feature	Categorical	–	Number of concomitant medical conditions	–	No
Coronary Artery Disease (CAD)	Feature	Binary	–	Presence of coronary artery disease	–	No
Hypothyroidism	Feature	Binary	–	Presence of an underactive thyroid gland	–	No
Hyperlipidemia	Feature	Binary	–	Presence of elevated blood lipid levels	–	No
Diabetes Mellitus (DM)	Feature	Binary	–	Presence of diabetes mellitus	–	No
Height	Feature	Integer	–	Height of the participant	–	No
Weight	Feature	Continuous	–	Body weight of the participant	–	No
Body Mass Index (BMI)	Feature	Continuous	–	Weight relative to height	–	No
Total Body Water (TBW)	Feature	Continuous	–	Total amount of water in the body	–	No

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Variable	Role	Type	Demo.	Description	Units	Missing
Extracellular Water (ECW)	Feature	Continuous	–	Amount of water outside body cells	–	No
Intracellular Water (ICW)	Feature	Continuous	–	Amount of water contained within body cells	–	No
Extracellular Fluid/Total Body Water (ECF/TBW)	Feature	Continuous	–	Ratio of extracellular fluid to total body water	–	No
Total Body Fat Ratio (TBFR)	Feature	Continuous	–	Proportion of total body mass consisting of fat	%	No
Lean Mass (LM)	Feature	Continuous	–	Proportion of lean body mass	%	No
Body Protein Content (Protein)	Feature	Continuous	–	Estimated proportion of body protein	%	No
Visceral Fat Rating (VFR)	Feature	Integer	–	Rating of visceral fat surrounding internal organs	–	No
Bone Mass (BM)	Feature	Continuous	–	Estimated bone mass	–	No
Muscle Mass (MM)	Feature	Continuous	–	Estimated amount of muscle mass	–	No
Obesity	Feature	Continuous	–	Estimated degree of excess adiposity	%	No
Total Fat Content (TFC)	Feature	Continuous	–	Estimated total amount of body fat	–	No
Visceral Fat Area (VFA)	Feature	Continuous	–	Estimated area of visceral adipose tissue	–	No
Visceral Muscle Area (VMA)	Feature	Continuous	–	Estimated visceral muscle measurement	kg	No
Hepatic Fat Accumulation (HFA)	Feature	Categorical	–	Degree of fat accumulation in the liver	–	No
Glucose	Feature	Continuous	–	Blood glucose level	–	No
Total Cholesterol (TC)	Feature	Continuous	–	Total blood cholesterol level	–	No
Low Density Lipoprotein (LDL)	Feature	Continuous	–	Low-density lipoprotein cholesterol level	–	No
High Density Lipoprotein (HDL)	Feature	Continuous	–	High-density lipoprotein cholesterol level	–	No
Triglyceride	Feature	Continuous	–	Triglyceride concentration in the blood	–	No
Aspartate Aminotransferase (AST)	Feature	Continuous	–	Liver enzyme measurement	–	No
Alanine Aminotransferase (ALT)	Feature	Continuous	–	Liver enzyme measurement	–	No
Alkaline Phosphatase (ALP)	Feature	Continuous	–	Enzyme associated primarily with liver and bone	–	No
Creatinine	Feature	Continuous	–	Indicator associated with kidney function	–	No
Glomerular Filtration Rate (GFR)	Feature	Continuous	–	Estimated kidney filtration rate	–	No
C-Reactive Protein (CRP)	Feature	Continuous	–	Marker associated with systemic inflammation	–	No

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Variable	Role	Type	Demo.	Description	Units	Missing
Hemoglobin (HGB)	Feature	Continuous	–	Concentration of oxygen-carrying haemoglobin	–	No
Vitamin D	Feature	Continuous	–	Blood vitamin D level	–	No

3.2.1 Categorical Variable Coding

Several categorical variables are represented by integer codes in the raw dataset. These values represent categories rather than continuous numerical quantities and therefore require appropriate treatment during preprocessing. The documented class labels are:

- **Gallstone Status:** 0 = gallstones present; 1 = gallstones absent.
- **Gender:** 0 = male; 1 = female.
- **Comorbidity:** 0 = no comorbidities; 1 = one comorbid condition; 2 = two comorbid conditions; 3 = three or more comorbid conditions.
- **Coronary Artery Disease:** 0 = no; 1 = yes.
- **Hypothyroidism:** 0 = no; 1 = yes.
- **Hyperlipidemia:** 0 = no; 1 = yes.
- **Diabetes Mellitus:** 0 = no; 1 = yes.
- **Hepatic Fat Accumulation (HFA):** 0 = no fat accumulation; 1 = Grade 1 (mild); 2 = Grade 2 (moderate); 3 = Grade 3 (severe); 4 = Grade 4 (very severe).

Although these categories are encoded numerically, they do not all have the same statistical interpretation. Binary variables such as diabetes and coronary artery disease can be represented as indicator variables, whereas *Comorbidity* and *Hepatic Fat Accumulation* are ordinal variables because their categories represent increasing levels of severity or burden. Their treatment during modelling will therefore be determined explicitly during the data-preparation stage.

3.3 Data exploration

Exploratory data analysis (EDA) was conducted on the raw dataset before any transformation or model training. The purpose was to understand the distribution of the target variable, examine the characteristics of the predictors, identify potential outliers or skewed variables, and investigate relationships that may influence subsequent data preparation and modelling.

3.3.1 Target Variable Distribution

Figure 5 shows the distribution of the target variable, *Gallstone Status*. Based on the documented class encoding, 0 represents the presence of gallstones and 1 represents the absence of gallstones.

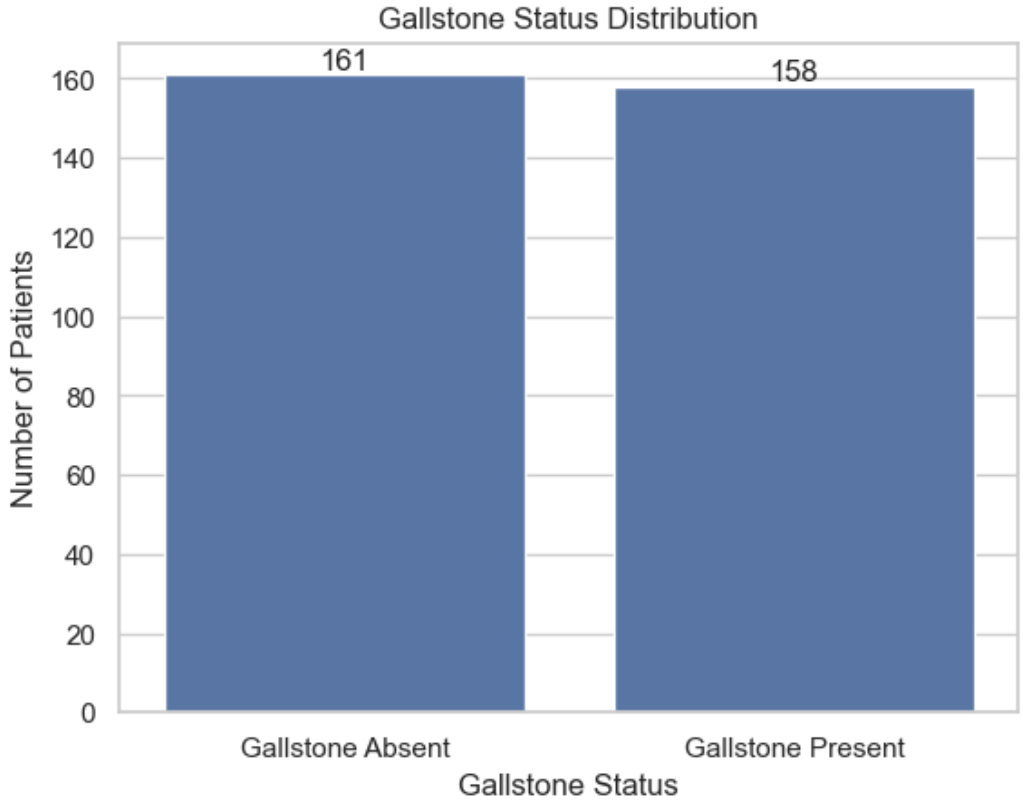


Figure 5: Distribution of Gallstone Status in the raw dataset.

The dataset contains 161 participants with gallstones and 158 participants without gallstones. The two classes therefore account for approximately 50.5% and 49.5% of the observations, respectively. This indicates that the classification problem is almost perfectly balanced. This is an important characteristic for the subsequent modelling stage. Severe class imbalance is unlikely to be a major source of bias, and resampling techniques such as oversampling or undersampling are not immediately required. Nevertheless, stratified data partitioning will be used when dividing the observations into training and testing sets to preserve the approximately equal class proportions. Model evaluation will also consider sensitivity, specificity, F1-score and ROC-AUC rather than accuracy alone.

3.3.2 Distribution of Continuous Features

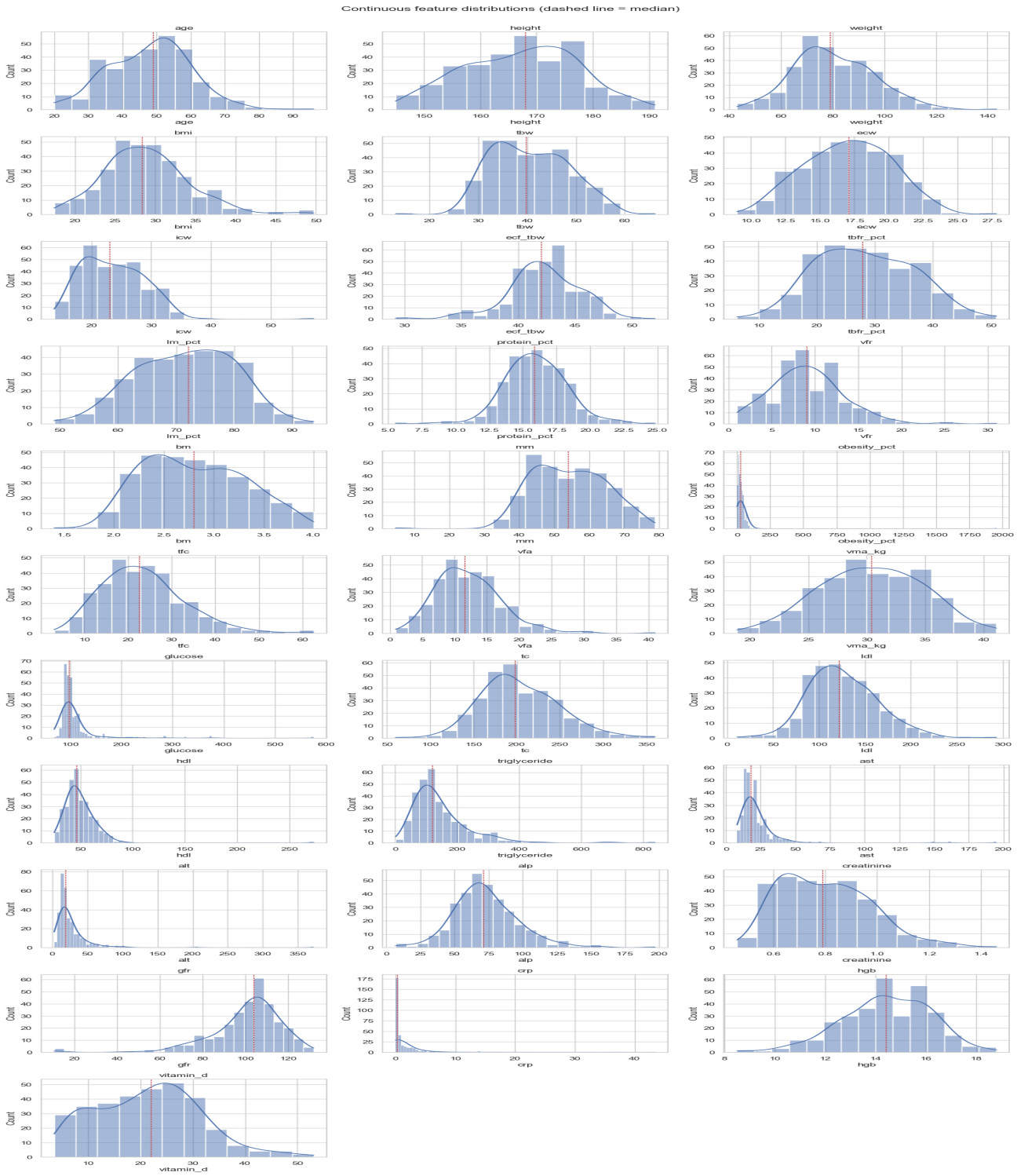


Figure 6: Distribution of continuous variables in the raw Gallstone dataset. The dashed vertical line represents the median.

Figure 6 presents the distributions of the continuous variables before any transformation or outlier treatment. The plots show considerable variation in distributional shape across the clinical and body-composition measurements.

Several variables, including age, height, BMI, extracellular water (ECW), lean mass, creatinine, haemoglobin, and visceral muscle area, display relatively smooth and approximately unimodal distributions. In contrast, variables such as glucose, triglyceride, AST, ALT, CRP, visceral fat rating (VFR), visceral fat area (VFA), and total fat content (TFC) show noticeable right-skewness, with most observations concentrated at lower values and a small

number of substantially larger measurements.

The distributions also reveal several extreme observations, particularly in metabolic and laboratory variables. These values require further assessment during the data-quality stage to determine whether they represent valid clinical measurements or potential data errors. They will not be removed automatically, as unusual values may contain clinically meaningful information relevant to gallstone risk.

Overall, this exploration indicates that the predictors differ substantially in both scale and distribution. Consequently, standardisation and possible transformations of strongly skewed variables will be considered during data preparation, depending on the modelling technique used. The next stage of the analysis compares these predictors with *Gallstone Status* to identify variables that may provide useful discriminatory information for the classification models.

3.3.3 Distribution of categorical features and Association with Gallstone Status

The categorical predictors were examined by comparing both the class counts and the proportion of participants with gallstones within each category. This provides an initial indication of whether particular categorical variables may contain useful discriminatory information for the prediction task.

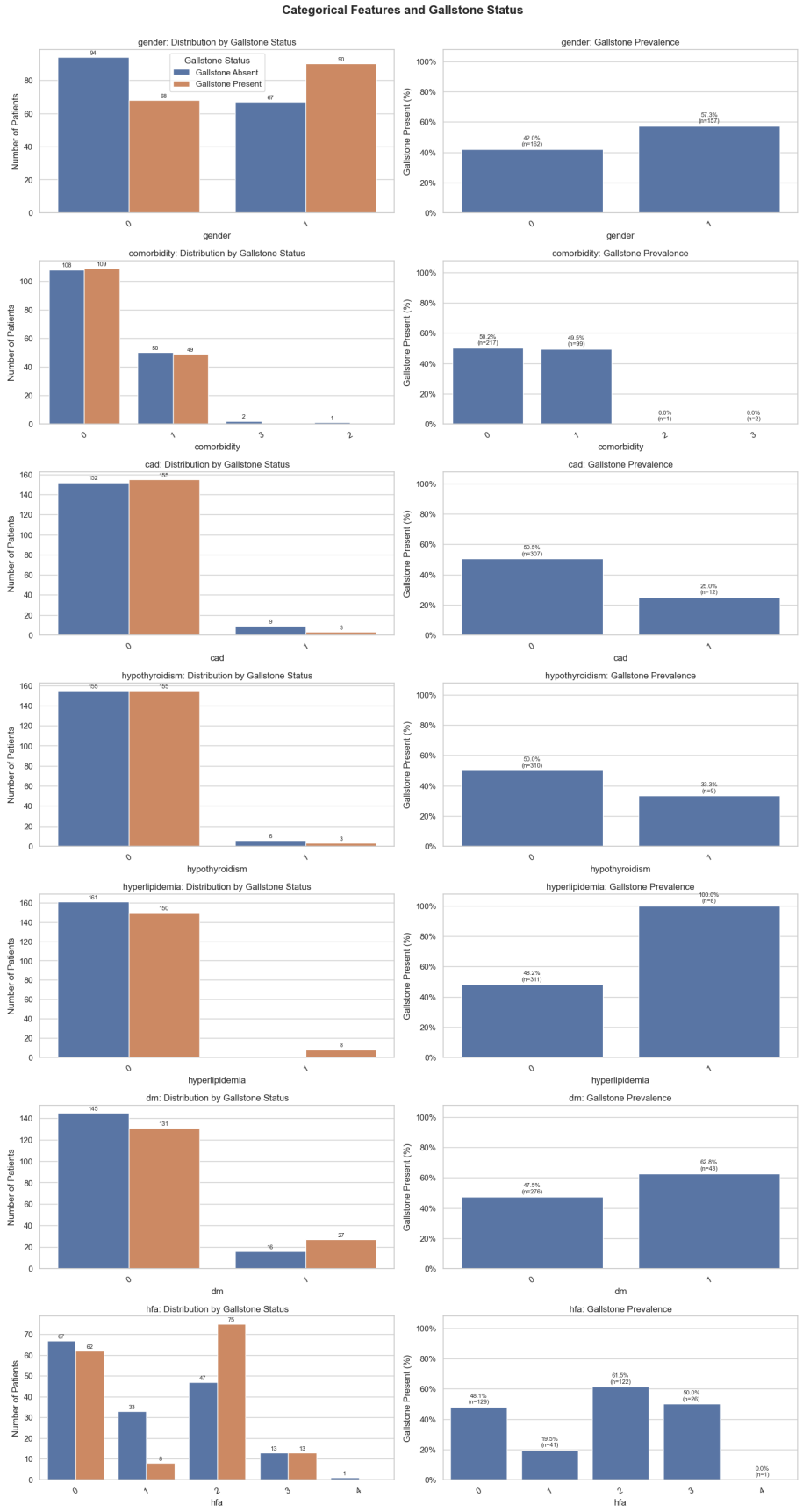


Figure 7: Distribution of categorical predictors by Gallstone Status and the corresponding gallstone prevalence within each category.

Figure 7 suggests that the predictive signal varies considerably across the categorical variables. Gender shows a moderate difference between categories: the gallstone prevalence is approximately 42.0% for category 0 and

57.3% for category 1, indicating that gender may provide some discriminatory information.

Diabetes mellitus (`dm`) also shows a potentially useful difference. Gallstone prevalence is approximately 47.5% among participants in category 0 compared with 62.8% in category 1. A similar pattern is visible for `hfa`, where prevalence varies across categories, most notably between category 1 (19.5%) and category 2 (61.5%). This suggests that these variables may contribute information useful for separating the two outcome classes.

In contrast, `comorbidity` shows little separation in the two major categories, with gallstone prevalence close to 50% in both. This indicates limited univariate discriminatory value for this variable. `cad` and `hypothyroidism` also show lower prevalence among category 1 than category 0, although the number of observations in their positive categories is small.

The strongest apparent difference is observed for `hyperlipidemia`, where all eight participants in category 1 have gallstones. However, this result should be interpreted cautiously because the category contains very few observations. Similar caution is required for rare levels of `comorbidity` and `hfa`, where prevalence estimates are based on only one or two participants and are therefore highly unstable.

Overall, the categorical EDA provides evidence of some class-related signal, particularly for gender, diabetes mellitus, and selected HFA categories. However, no single categorical predictor provides a sufficiently strong or stable separation between participants with and without gallstones. The results therefore support retaining these variables for multivariable modelling, where their combined effects and interactions with continuous clinical and body-composition measurements can be evaluated.

3.3.4 Correlation Analysis

Figure 8 presents the Spearman correlationmatrix for the dataset. Spearman correlation was selected because several continuous variables are skewed and contain extreme observations.

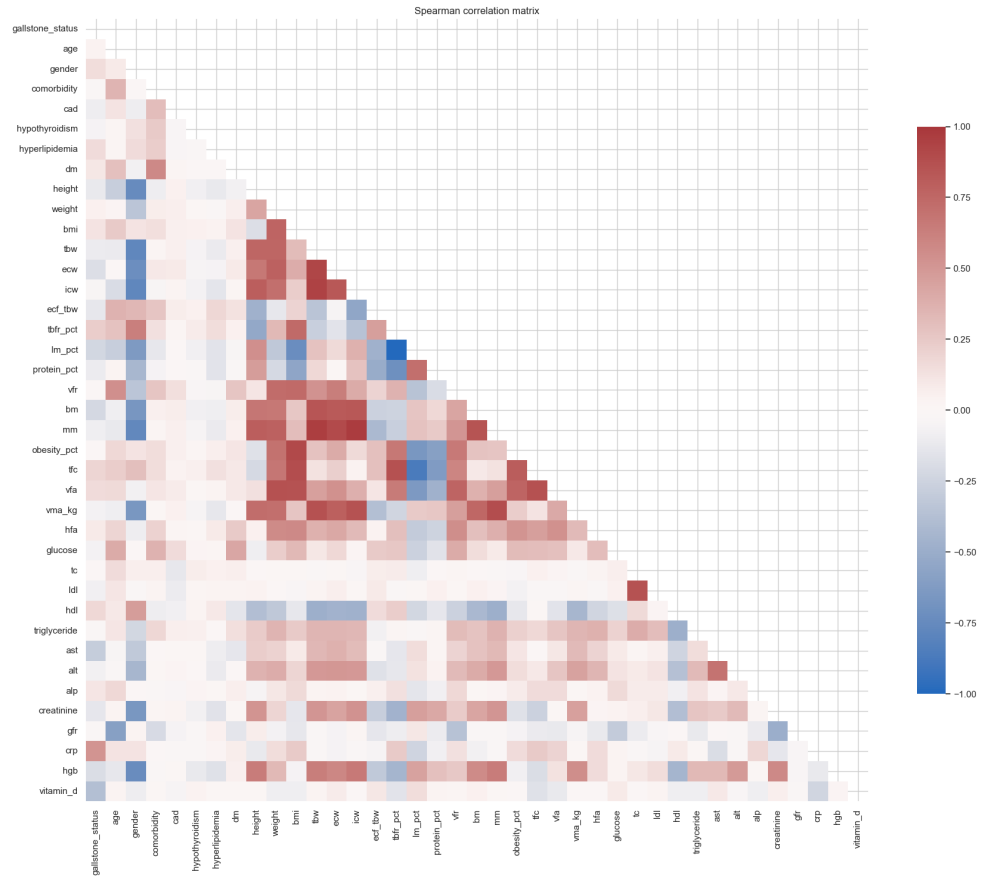


Figure 8: Spearman correlation matrix for the Gallstone dataset visualization.

The matrix shows several groups of strongly related predictors. In particular, body-composition variables such as weight, BMI, TBW, ECW, ICW, muscle mass, body-fat measures, and visceral-fat measures show substantial correlation. Laboratory variables also contain related groups, including TC-LDL, AST-ALT, and the inverse relationship between creatinine and GFR. These patterns indicate potential redundancy and multicollinearity among some predictors.

In contrast, individual correlations with *Gallstone Status* are generally weaker. This suggests that gallstone status is unlikely to be explained by a single predictor and supports the use of multivariable classification models that combine demographic, body-composition, and laboratory information.

These findings will guide the data-preparation stage. Highly correlated variables will be reviewed during feature selection, particularly for logistic regression, while tree-based models will also be considered because they are less sensitive to multicollinearity.

3.3.5 Structure Among Body-Composition Variables

Some body-composition variables, including total body water (TBW), bone mass (BM), and muscle mass (MM), appear to have broader or mildly multi-peaked distributions. This may indicate underlying differences between participant subgroups, for example differences in sex, body size, or body composition. At this stage, however, the univariate plots alone are insufficient to determine the cause. Furthermore, several predictors measure related physiological concepts. For example, BMI, TBFR, TFC, VFR and VFA all contain information relating to body fat or adiposity, while TBW, ECW and ICW describe different components of body watxer. Consequently, correlation analysis will be conducted in the next stage to determine whether some predictors contain highly overlapping information. Highly correlated variables will be reviewed before modelling to reduce unnecessary redundancy and potential multicollinearity, particularly for regression-based classifiers.

Overall, the EDA provides the foundation for the next data-preparation phase by identifying class balance, distributional characteristics, outliers, potentially informative predictors, and redundant variables. These findings will subsequently guide preprocessing, feature selection, model selection, and evaluation.

3.4 Data Quality

Data quality was assessed before any transformation or model development to ensure that errors in the raw data would not be carried into the predictive models. The verification focused on missing values, duplicate records, categorical coding, numerical ranges, and potential outliers.

3.4.1 Missing Values and Completeness

The dataset documentation reports no missing values, and this was confirmed during the initial inspection of the dataset. Therefore, no imputation was required at this stage. Nevertheless, missing-value checks will remain part of the preprocessing workflow to ensure that any values introduced during transformation or partitioning are detected before modelling.

3.4.2 Potential Outliers and Data Quality Issues

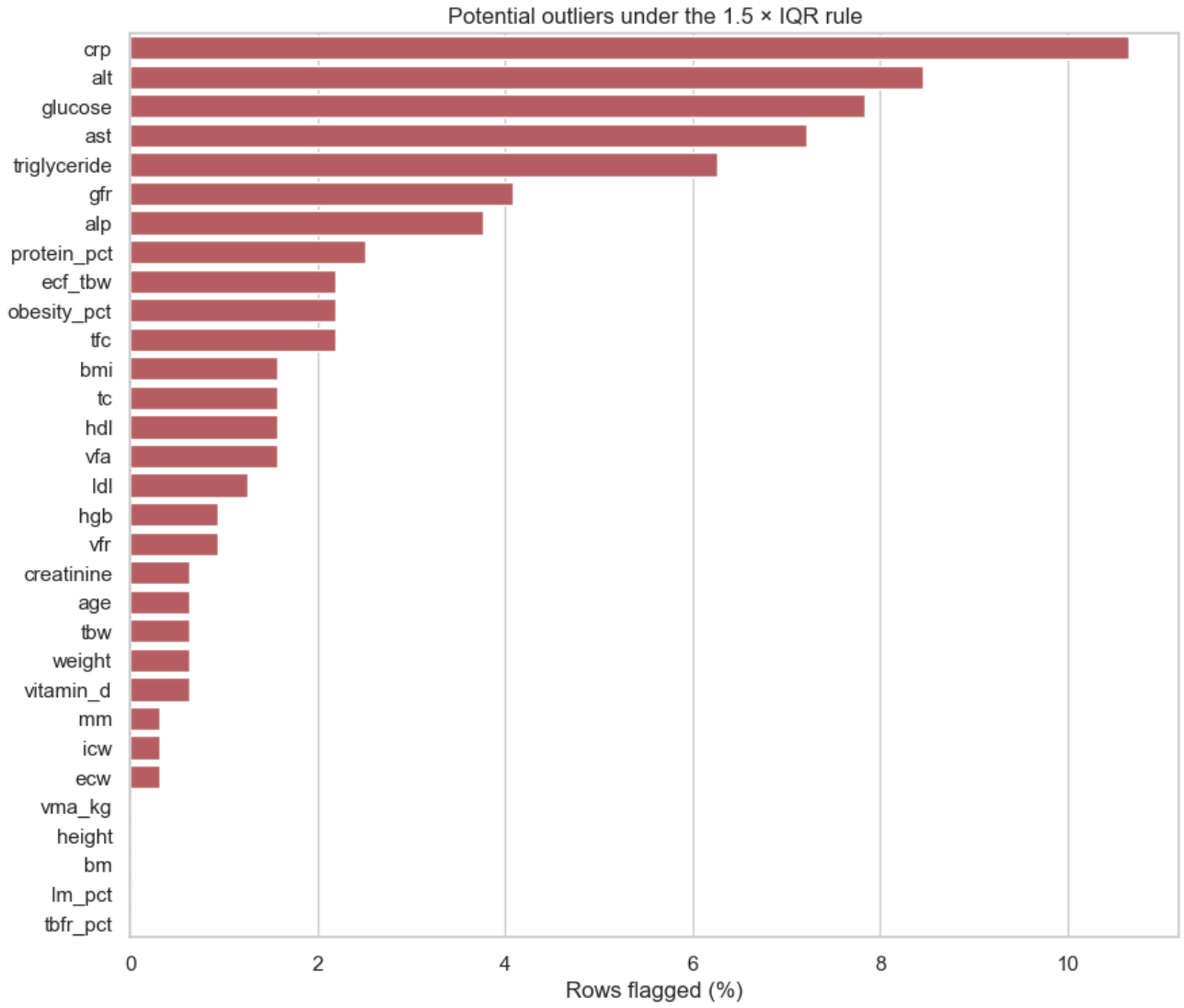


Figure 9: Outlier analysis using the 1.5×IQR method.

Figure 9 identifies several observations that require further investigation. Most notably, the *Obesity* variable contains an extreme value of approximately 2000, which is substantially larger than the remaining observations. Given the magnitude of this difference, the value should be verified against the original dataset and variable definition to determine whether it represents a valid measurement, a unit inconsistency, or a data-entry error.

Potential upper-tail outliers are also observed in variables including glucose, HDL, triglyceride, AST, ALT, CRP, and intracellular water (ICW). However, observations identified by the 1.5×IQR rule should not automatically be treated as erroneous, particularly in clinical data where extreme measurements may represent genuine patient conditions. Therefore, these observations will be reviewed during data preparation before any correction, transformation, or removal is applied.

Valid extreme observations will be retained where appropriate to preserve clinically meaningful variation in the dataset. Only values supported by evidence of coding, measurement, or data-entry errors will be considered for correction or exclusion. This approach reduces the risk of removing informative clinical cases and introducing bias into subsequent predictive models.

3.4.3 Dimensionality Reduction Visualisation and Data Understanding

Dimensionality-reduction techniques were used to obtain a more compact view of the relationships among the continuous measurements and to assess whether participants with and without gallstones exhibit distinct multivariate patterns. Two complementary approaches were considered: Principal Component Analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE). PCA provides a linear representation of the dominant sources of variance in the data, whereas t-SNE is a nonlinear technique designed to preserve local neighbourhood relationships in a low-dimensional representation (van der Maaten & Hinton, 2008).

Principal Component Analysis PCA was first applied to the continuous measurements to examine the main directions of variation in the dataset. The first principal component (PC1) explained 27.7% of the total variance, while the second principal component (PC2) explained a further 20.0%. Together, the first two components therefore accounted for approximately 47.7% of the variance in the continuous feature space.

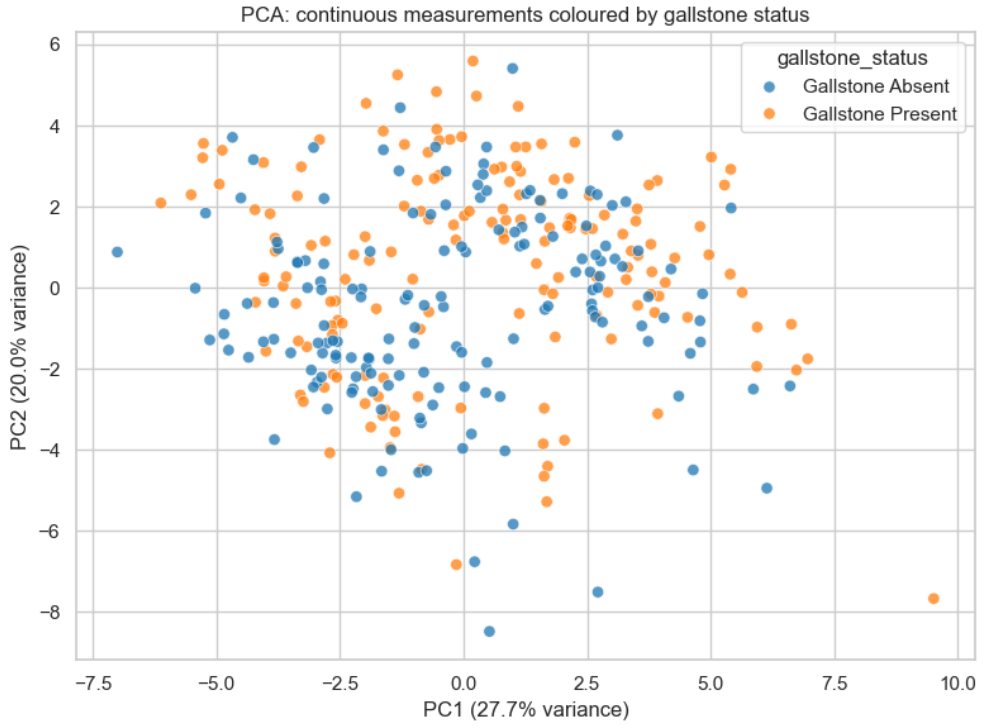


Figure 10: Two-dimensional PCA projection of the continuous measurements, coloured by Gallstone Status. PC1 and PC2 explain 27.7% and 20.0% of the total variance, respectively.

As shown in Figure 10, observations from the *Gallstone Present* and *Gallstone Absent* groups overlap substantially in the first two principal components. No distinct linear separation between the two outcome groups is apparent. This indicates that the largest sources of variation in the continuous measurements are not themselves equivalent to the variation that separates gallstone status.

Nevertheless, the projection reveals meaningful structure within the measurements. Participants are distributed across several regions of the PCA space, suggesting heterogeneity in body-composition, anthropometric, and related clinical characteristics. A small number of observations also lie relatively far from the central concentration of the data. These observations are consistent with the unusual or extreme measurements identified during the earlier univariate exploration and were retained for further investigation rather than being removed solely on the basis of their position in the PCA projection.

To understand which variables contribute most strongly to the two principal components, the corresponding PCA loadings were also examined.

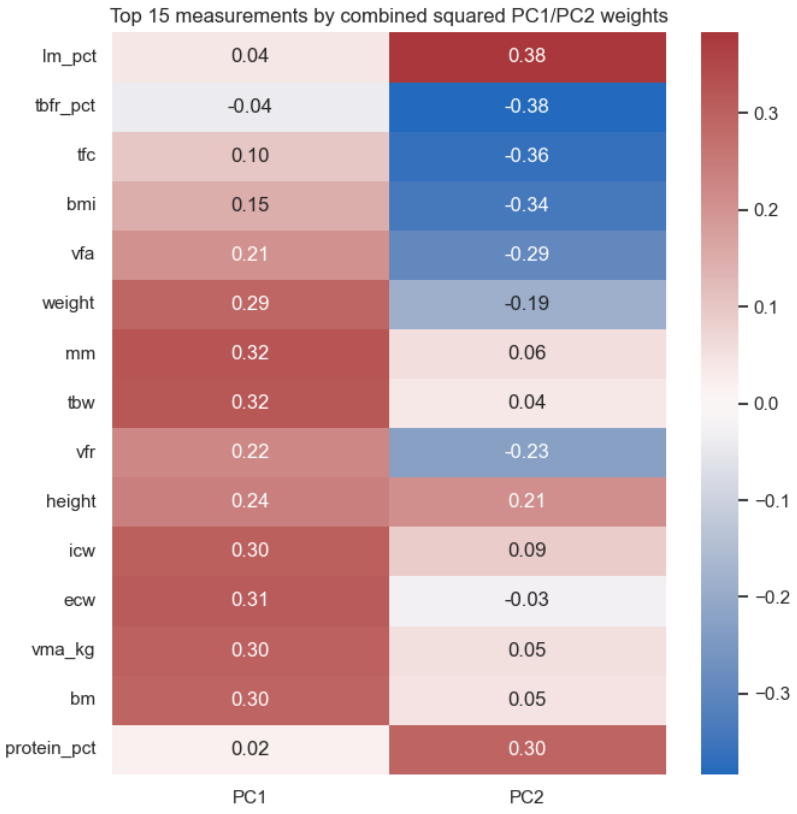


Figure 11: Loadings of the 15 continuous measurements with the largest combined squared contributions to PC1 and PC2. Positive and negative values indicate the direction of association with each principal component.

Figure 11 shows that PC1 is influenced primarily by several body-size and body-composition measurements. Muscle mass (`mm`), total body water (`tbw`), extracellular water (`ecw`), intracellular water (`icw`), visceral muscle area (`vma_kg`), bone mass (`bm`), weight, and height all have positive loadings on PC1. This suggests that PC1 broadly represents variation associated with overall body size and lean/body-composition characteristics.

PC2 captures a different contrast. Lean-mass percentage (`lm_pct`) and protein percentage (`protein_pct`) have strong positive loadings, whereas total body-fat percentage (`tbfr_pct`), total fat content (`tfc`), BMI, visceral fat area (`vfa`), and visceral fat rating (`vfr`) have negative loadings. PC2 can therefore be interpreted as reflecting a contrast between lean-body composition and fat-related measurements. These loading patterns provide additional insight into the correlations among the original measurements and help explain the structure observed in the PCA projection.

t-SNE Visualisation Because PCA represents only linear combinations of the original variables, a nonlinear t-SNE analysis was additionally performed to examine whether local neighbourhood structures could reveal patterns that were not visible in the PCA projection. Multiple perplexity values were evaluated rather than relying on a single t-SNE configuration.

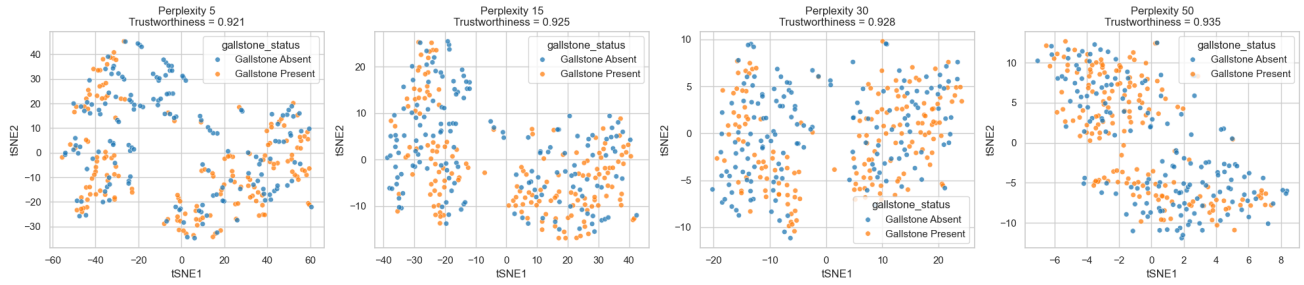


Figure 12: Two-dimensional t-SNE projections of the continuous measurements using perplexity values of 5, 15, 30, and 50. Points are coloured according to Gallstone Status. Trustworthiness scores are reported for each configuration.

Figure 12 shows that the overall appearance of the t-SNE representation changes as the perplexity parameter increases. Lower perplexity values, particularly 5, emphasise very local neighbourhoods and produce a more fragmented representation. At larger perplexities, the observations form broader and more coherent structures.

The trustworthiness score increased from 0.921 at perplexity 5 to 0.935 at perplexity 50, indicating that the higher-perplexity representations preserved the original local neighbourhood relationships slightly more effectively. However, the purpose of this metric is to assess preservation of neighbourhood structure rather than separation of the target classes.

Across all perplexity settings, participants with and without gallstones remain substantially intermixed. Although some local regions contain a greater concentration of one class than the other, there is no stable class-specific cluster or simple visual boundary that consistently separates *Gallstone Present* from *Gallstone Absent*. The same general conclusion is therefore observed in both the linear PCA representation and the nonlinear t-SNE representations.

These findings suggest that gallstone status is unlikely to be explained by a single dominant measurement or by a simple low-dimensional boundary. Instead, the predictive information may be distributed across several demographic, body-composition, metabolic, and biochemical variables and may involve interactions or nonlinear relationships among them. This observation motivates the subsequent comparison of multivariable classification models.

Importantly, both PCA and t-SNE were used here as exploratory tools for data understanding rather than as measures of predictive performance. Visual overlap between the classes does not imply that the target is unpredictable, since discriminative information may exist in dimensions that are not represented by the first two principal components or by the two-dimensional t-SNE embedding. Model performance is therefore assessed separately using the validation and test partitions.

4 Data Preparation

The purpose of the data-preparation stage was to transform the raw Gallstone dataset into a consistent, traceable, and modelling-ready dataset. Decisions made during this stage were informed by the business objective, findings from the exploratory analysis, data-quality checks, and the technical requirements of the modelling pipeline.

All data-preparation procedures were implemented using an open-source Python workflow. The main libraries used were `pandas` and `NumPy` for data manipulation and validation, `SciPy` for statistical transformations, `scikit-learn` for data partitioning and preprocessing, and `Matplotlib` and `Seaborn` for diagnostic visualisation.

Using Python rather than a graphical modelling environment provides a script-based and reproducible workflow in which each preprocessing operation can be inspected, rerun, and consistently applied to subsequent modelling experiments.

4.1 Data Selection

The modelling dataset contains *Gallstone Status* as the target variable and the available demographic, comorbidity, body-composition, and laboratory measurements as candidate predictors. Variables were initially retained rather than removed solely on the basis of weak univariate relationships because the exploratory analysis suggested that gallstone status may depend on combinations of multiple patient characteristics.

Data selection was performed programmatically using Python. The dataset was loaded into a `pandas DataFrame`, after which column names, data types, unique values, distributions, and summary statistics were inspected. Candidate predictors were retained when they had plausible clinical relevance and sufficient data quality for subsequent modelling.

Variables containing unusual values, strong skewness, or high correlations with other predictors were not immediately removed. Instead, these characteristics were investigated during exploratory analysis and preprocessing. In particular, highly correlated predictors were reviewed during feature selection because multicollinearity may influence coefficient stability in models such as logistic regression.

No patient-identifying information or variables without potential predictive value were required for model development. Identifier fields used for data traceability were retained during data processing but excluded from the feature matrix supplied to the predictive models.

4.2 Data Cleaning

Data cleaning was guided by the issues identified during the earlier data-quality assessment. The Python workflow systematically evaluated missing values, duplicated records, invalid categorical codes, numerical ranges, data types, and potential extreme observations.

Completeness was assessed using functions such as `DataFrame.isna()`, while duplicate observations and field-level consistency were examined using standard `pandas` operations. Summary statistics and value-frequency tables were also generated to identify unexpected values and inconsistencies.

No missing values were identified in the original dataset; therefore, no imputation was required before modelling. Nevertheless, missing-value checks were retained as part of the preprocessing pipeline so that the workflow can identify incomplete observations if the dataset is modified or extended in future iterations.

Categorical variables were checked against their documented coding ranges. Binary and ordinal fields stored numerically were treated according to their clinical meaning rather than automatically interpreted as continuous measurements.

Potential outliers were identified during exploratory analysis using the 1.5×IQR rule:

$$\text{Lower Bound} = Q_1 - 1.5(Q_3 - Q_1),$$

$$\text{Upper Bound} = Q_3 + 1.5(Q_3 - Q_1).$$

Observations outside these limits were reviewed rather than automatically removed. This distinction is important for clinical data because statistically extreme observations may nevertheless correspond to genuine patient measurements.

In particular, the unusually large observation identified in the *Obesity* variable was reviewed as a possible data-quality issue. Similarly, extreme observations in glucose, triglyceride, AST, ALT, CRP, and other laboratory measurements were retained when they represented plausible clinical values. Records would only be corrected or excluded if there were sufficient evidence of a coding, unit, or data-entry error.

Column names were also standardised programmatically to remove unnecessary spaces and special characters and to provide consistent variable names throughout the analysis.

4.2.1 Cleaning and Sanity-Checking Workflow

The data-cleaning process was implemented as a reproducible sequence of Python operations. The main stages were:

1. verify the number of observations and variables;
2. inspect data types and expected variable roles;
3. identify missing and blank values;
4. detect duplicate observations;
5. validate categorical coding ranges;
6. examine descriptive statistics for continuous measurements;
7. identify potential extreme observations;
8. standardise column names and field representations; and
9. confirm the dimensions and integrity of the resulting analytical dataset.

This workflow provided explicit quality-control checkpoints while preserving the original observations whenever no clear data-quality issue was present.

4.2.2 Skewness Assessment and Transformation of Continuous Features

The distributions of the continuous predictors were further examined using sample skewness to identify variables with substantial departures from symmetry. Absolute skewness values greater than or equal to 1 were treated as an indication of strong skewness and were selected for further transformation assessment.

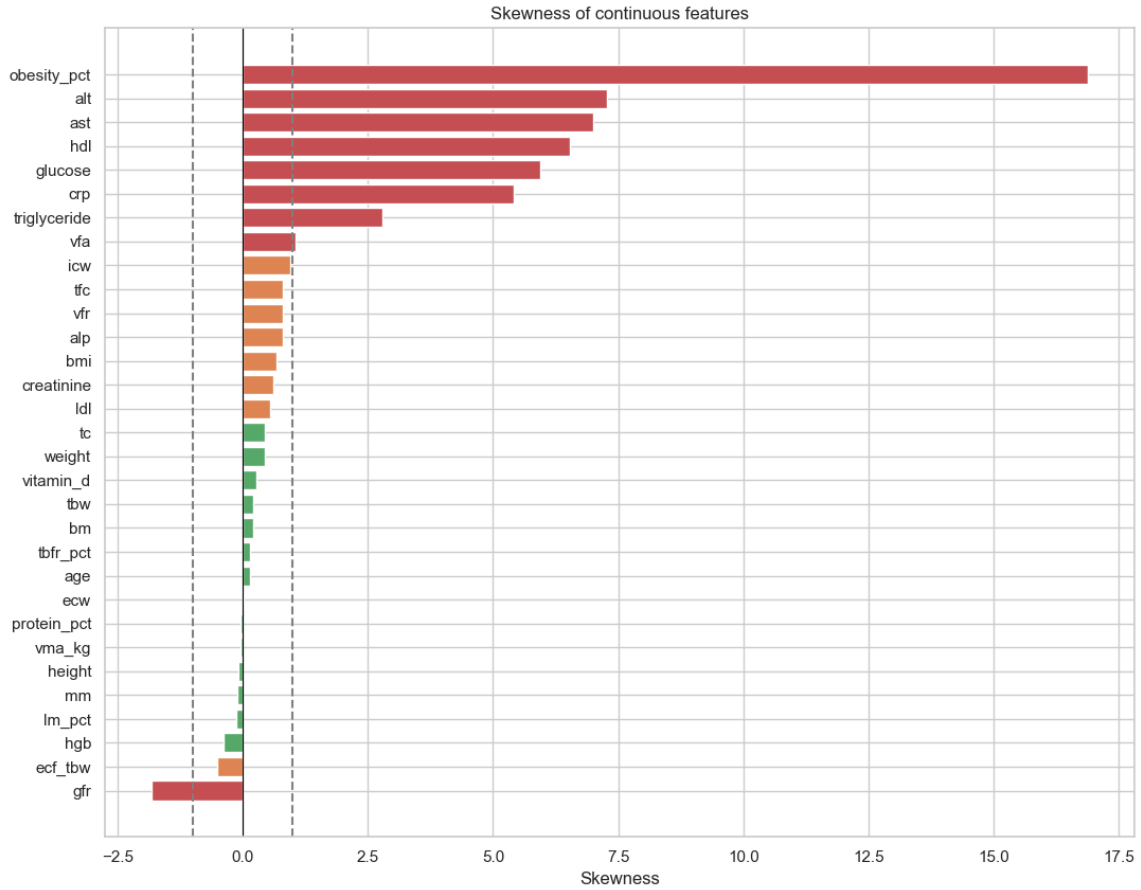


Figure 13: Skewness of the continuous variables. The dashed vertical lines indicate the threshold of |skewness| = 1 used to identify strongly skewed variables.

Figure 13 shows that several continuous measurements were strongly right-skewed. The most extreme case was `obesity_pct`, with a skewness of approximately 16.87, followed by ALT (7.28), AST (6.99), HDL (6.53), glucose (5.94), and CRP (5.41). Triglyceride was also strongly positively skewed (2.79), while GFR exhibited substantial negative skewness (-1.81). VFA showed a smaller but still notable positive skewness of approximately 1.06. In contrast, most of the remaining continuous measurements were relatively close to symmetric.

These results are consistent with the earlier outlier analysis. In particular, the extreme skewness of `obesity_pct` appears to be strongly influenced by the unusually large observation identified during data-quality assessment. Similarly, the long right tails observed for ALT, AST, glucose, CRP, and triglyceride indicate that a small number of participants have measurements considerably larger than the majority of the sample. These observations were not removed solely because they increased skewness, since extreme values may represent clinically plausible measurements.

4.3 Data Construction

The data-construction stage focused on producing a consistent analytical dataset while preserving the original source data. No additional derived clinical variables were introduced unless they had a clear analytical justification.

Three logical versions of the dataset were maintained:

- **Raw data:** the original dataset retained without modification;
- **Cleaned data:** the dataset after validation of field names, data types, categorical values, ranges, duplicates, and potential data-quality issues;

- **Model-ready data:** the final analytical dataset containing *Gallstone Status* and the predictor variables used by the Python modelling pipeline.

This separation ensures that preprocessing operations do not overwrite the source dataset and allows each transformation to be reproduced from the raw data through the Python scripts.

4.3.1 Data Partitioning

Before feature selection and model development, two complementary data-splitting and evaluation strategies were adopted to assess model performance and robustness: a stratified 70–15–15 holdout split and a *Leave-One-Patient-Out* (LOPO) cross-validation procedure.

Strategy 1: Stratified 70–15–15 Holdout Split For the primary holdout evaluation, the model-ready dataset was partitioned using the `scikit-learn` library with stratification based on *Gallstone Status*. The following proportions were used:

- **70% Training set:** used for preprocessing decisions, feature selection, and model fitting;
- **15% Validation set:** used for model comparison, hyperparameter tuning, threshold analysis, and model-selection decisions;
- **15% Test set:** retained for the final evaluation of the selected modelling approach on previously unseen observations.

Stratification was used to preserve the approximately balanced distribution of participants with and without gallstones across the three subsets. This was particularly important given the relatively small dataset because substantial differences in class proportions between partitions could otherwise introduce additional variability into the evaluation results.

The partitioning process can be represented conceptually as follows:

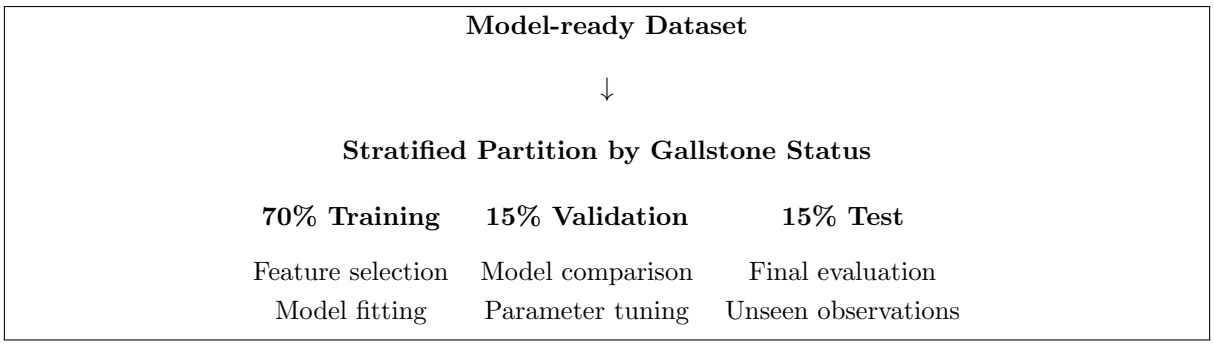


Figure 14: Training, validation, and testing strategy used in the Python modelling workflow.

Feature-selection decisions and model fitting were performed using only the training partition. The validation partition was subsequently used to compare candidate models and tune modelling choices. The final test partition remained separate from these stages and was used only after the modelling strategy had been established.

Strategy 2: Leave-One-Patient-Out Evaluation A second evaluation strategy, referred to as *Leave-One-Patient-Out* (LOPO), was used to provide an additional assessment of model robustness on unseen patients. LOPO corresponds to leave-one-out cross-validation at the patient level. During each iteration, one patient was held out for evaluation while all remaining patients were used for model training.

For a dataset containing N patients, the procedure generates N model-fitting iterations:

$$\mathcal{D}_{\text{train}}^{(i)} = \mathcal{D} \setminus \{P_i\}, \qquad \mathcal{D}_{\text{test}}^{(i)} = \{P_i\},$$

where P_i denotes the patient excluded from the training data during iteration i.

This process was repeated until every patient had served once as the unseen evaluation case. The predictions generated across all iterations were then combined to calculate the overall evaluation metrics.

The LOPO strategy differs from the holdout approach because almost the entire dataset is available for model fitting during each iteration. This is particularly useful for the Gallstone dataset because of its relatively small sample size and provides an assessment of whether the model can generalise to an individual patient who was completely excluded from the corresponding training process.

The two strategies were evaluated separately. The 70–15–15 split provides a conventional development framework with independent training, validation, and test partitions, while LOPO evaluates model stability and patient-level generalisation across the complete dataset. Using both approaches provides a more comprehensive assessment of predictive performance than relying on a single data split alone.

These separation reduces the risk of information leakage and provides a more independent estimate of how well the resulting model is expected to generalise to previously unseen patient observations.

Any preprocessing operation that required parameters to be estimated from the data was fitted using the training partition only and subsequently applied to the validation and test partitions. This principle applies, for example, to feature scaling, statistical transformations, and model-based feature selection.

4.4 Data Integration

Although the Gallstone dataset was originally obtained as a single public dataset, an explicit integration exercise was conducted to demonstrate how logically separate healthcare data sources can be combined into a unified analytical table.

The original dataset was preserved unchanged and divided into two logical CSV sources. The first file, `gallstone_demographic` contains 319 records and 9 fields describing demographic and clinical characteristics. The second file, `gallstone_biological_b` contains the same 319 participants and 33 fields describing biological, laboratory, and body-composition measurements.

A unique `record_id`, ranging from 1 to 319, was included in both tables as the one-to-one integration key. Both tables also contained *Gallstone Status* so that target consistency could be verified before integration. However, *Gallstone Status* was not used as a merge key because it does not uniquely identify an individual participant.

Integration was implemented using the `pandas.merge()` operation with `record_id` as the matching field. The merge was configured as a one-to-one operation so that duplicate identifiers or unexpected many-to-one relationships could be detected explicitly.

Post-integration quality checks verified that all 319 participant records were retained and that no unmatched identifiers or additional missing values were introduced by the merge.

The duplicated *Gallstone Status* fields from the two logical data sources were compared before one copy was removed from the final analytical table. The `record_id` field was retained for traceability and quality control but excluded from the predictor set used for model development.

This integration exercise represents a simplified version of a realistic healthcare environment in which demographic and clinical data may be stored separately from laboratory or body-composition measurements. The Python workflow demonstrates how these sources can be integrated while maintaining record integrity and preventing inappropriate use of the target variable as an integration key.

4.5 Data Reformatting

The original Gallstone dataset was provided in Microsoft Excel format (`.xlsx`). The file was loaded programmatically using `pandas.read_excel()` and subsequently exported to comma-separated values (`.csv`) where appropriate for intermediate storage and reproducible processing.

CSV was selected as an intermediate format because it provides a simple, platform-independent representation of tabular data and avoids spreadsheet formatting that is irrelevant to the modelling process. The use of CSV was not a modelling requirement, however, because the Python pipeline can also operate directly on Excel or other structured tabular formats.

During reformatting, variable names were standardised and unnecessary spreadsheet-specific formatting was removed. The resulting fields were then classified according to their analytical meaning.

Continuous clinical and body-composition measurements, such as BMI, glucose, triglyceride, weight, and total body water, were represented as numerical variables.

Binary variables such as Coronary Artery Disease, Hypothyroidism, Hyperlipidemia, and Diabetes Mellitus were stored as categorical or binary indicators rather than being interpreted as continuous measurements.

Variables representing ordered clinical levels were treated as ordinal variables when their coding reflected an increasing degree of severity or disease burden.

The target variable, *Gallstone Status*, was represented internally as a binary label for model training while human-readable labels (*Gallstone Absent* and *Gallstone Present*) were used in tables and visualisations.

The final model-ready dataset therefore contained consistent field names, validated data types, clearly defined variable roles, and a reproducible schema that could be used directly by the subsequent Python-based feature engineering and modelling pipeline.

4.6 Reproducible Open-Source Preparation Pipeline

The complete preparation process was implemented as a script-based open-source workflow. At a high level, the pipeline consisted of:

1. loading the source data using `pandas`;
2. standardising variable names and data types;
3. validating missing values, duplicates, coding ranges, and observations;
4. performing exploratory outlier and distribution checks;
5. integrating logically separated datasets using `pandas.merge()`;
6. constructing the final predictor and target datasets;
7. creating stratified training, validation, and test partitions using `scikit-learn`;
8. fitting any required preprocessing operations using the training subset only; and
9. passing the resulting datasets to the machine-learning modelling pipeline.

The use of an open-source Python stack provides several advantages for this project. Each operation is explicitly defined in code, preprocessing decisions can be version-controlled, experiments can be reproduced using fixed random seeds, and identical transformations can be consistently applied across multiple candidate models.

The resulting workflow therefore provides a transparent and reproducible transition from the original Gallstone dataset to the datasets used for model training, validation, and final evaluation.

4.7 Data Reduction and Projection

4.7.1 Feature Selection and Horizontal Data Reduction

Horizontal data reduction was performed to retain predictors that provide useful information for predicting *Gallstone Status*, while reducing unnecessary or weakly informative inputs. This was particularly relevant because the exploratory analysis identified substantial correlation among several body-composition and laboratory measurements.

Random Forest Feature Importance A Random Forest classifier was used as an exploratory feature-ranking method. Random Forests combine multiple decision trees constructed from bootstrap samples and random subsets of predictors, allowing nonlinear relationships and interactions among variables to be captured without requiring strong parametric assumptions (Breiman, 2001).

Feature importance was calculated using the impurity-based importance measure provided by the RandomForestClassifier implementation in `scikit-learn`[1].

For each predictor, the importance reflects its normalised contribution to the total reduction in the splitting criterion across the trees in the forest. The reported values were averaged across the repeated holdout splits, with the error bars indicating variability in importance across the different random seeds.

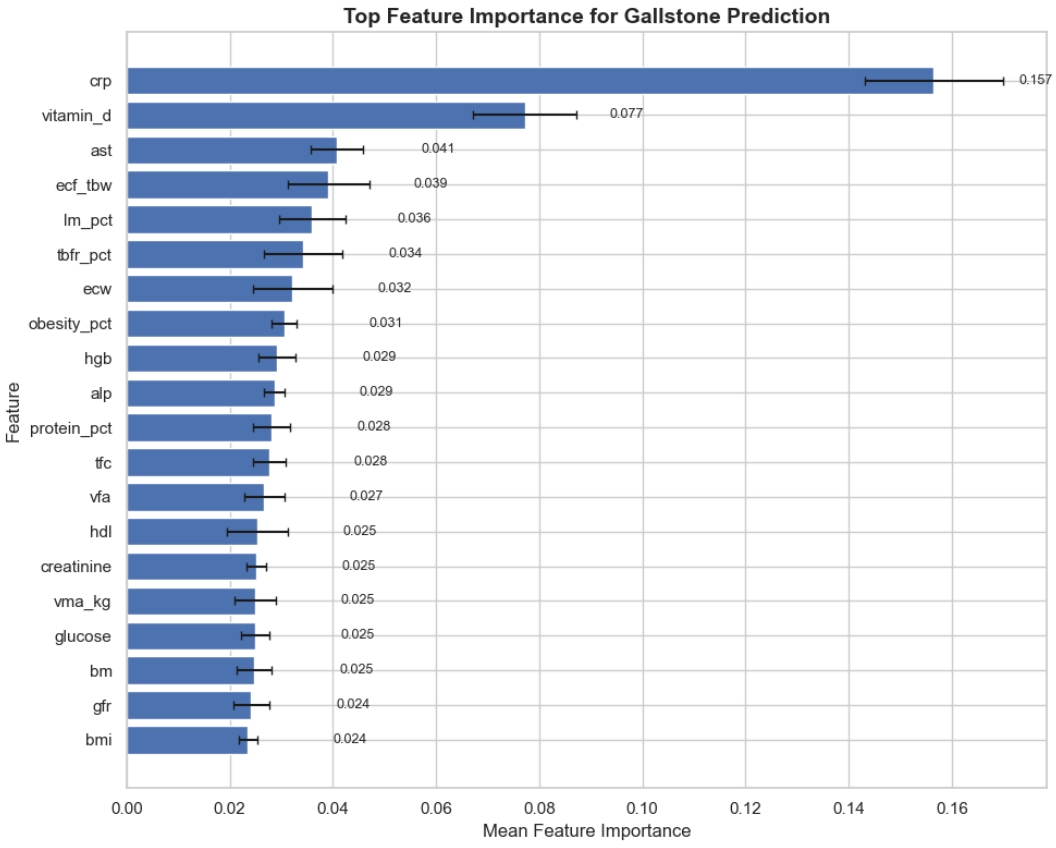


Figure 15: Mean Random Forest feature importance for Gallstone Status prediction across repeated stratified holdout splits. Error bars represent the variability of feature importance across random seeds.

As shown in Figure 15, CRP was the most influential predictor, with a mean importance of approximately 0.157. This was followed by Vitamin D (0.077), AST (0.041), ECF/TBW (0.039), lean-mass percentage (0.036), and total body-fat percentage (0.034). ECW, `obesity_pct`, haemoglobin, ALP, protein percentage, total fat content,

[1]The implementation and definition of `feature_importances_` are documented in the scikit-learn `RandomForestClassifier` API: https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html.

VFA, HDL, creatinine, glucose, GFR, and BMI also contributed to the model, but with smaller individual importance scores.

The ranking suggests that the predictive information is distributed across multiple clinical domains rather than being dominated by a single type of measurement. In particular, inflammatory and biochemical variables such as CRP, AST, and Vitamin D appear alongside body-composition measurements such as ECF/TBW, lean-mass percentage, total body-fat percentage, ECW, and VFA. This supports the earlier exploratory findings that Gallstone Status is associated with a multivariable pattern rather than a simple separation based on one predictor.

The Random Forest ranking was used as an exploratory assessment of predictor relevance rather than as the sole criterion for feature removal. Impurity-based importance can favour variables that provide more potential split points, and importance values should therefore be interpreted as model-specific indicators rather than direct measures of clinical effect (Pedregosa et al., 2011). Final feature-selection decisions were evaluated separately using recursive feature elimination and predictive performance across the validation procedures.

Feature Selection After Transformation Following the statistical transformation stage, feature selection was performed using the open-source Python modelling stack. Both the original predictors and the transformed representations were included as candidate inputs so that the usefulness of each representation could be evaluated empirically rather than assuming that transformation would automatically improve prediction.

Horizontal data reduction was performed using Recursive Feature Elimination (RFE) with a CART decision-tree classifier as the underlying estimator. CART provides a nonlinear feature-ranking mechanism based on reductions in node impurity, while RFE recursively removes the least informative predictors and re-fits the estimator on the remaining feature set. This procedure therefore reduces the dimensionality of the modelling dataset while retaining variables that contribute most strongly to the separation of *Gallstone Present* and *Gallstone Absent* cases.

Feature selection was performed within the training partition for each random seed to avoid using information from the validation or test data. The resulting feature rankings were then compared across repeated holdout splits to assess the stability of the selected predictors.

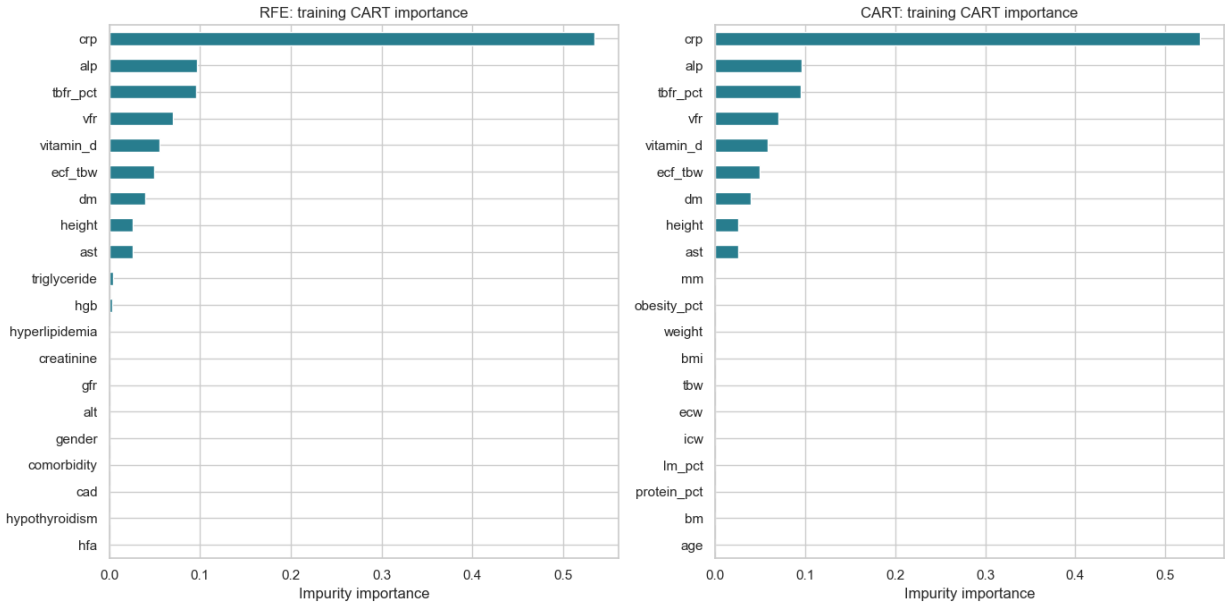


Figure 16: CART-based feature importance used within the recursive feature elimination procedure. Predictor importance was estimated using the training data only.

As shown in Figure 16, CRP was the dominant predictor in the CART-based ranking. ALP and total body-fat percentage (`tbfr_pct`) formed the next most influential group, followed by visceral fat rating (`vfr`), Vitamin

D, ECF/TBW, and diabetes mellitus (dm). Several additional variables, including height and AST, contributed more weakly to the fitted tree.

The dominance of CRP was also visible in the individual CART structures. In the two example training splits shown in Figures 17 and 18, CRP was selected as the root split, while subsequent branches incorporated different combinations of body-composition and clinical variables such as bone mass, ECF/TBW, total body-fat percentage, Vitamin D, VFR, BMI, and ALP.

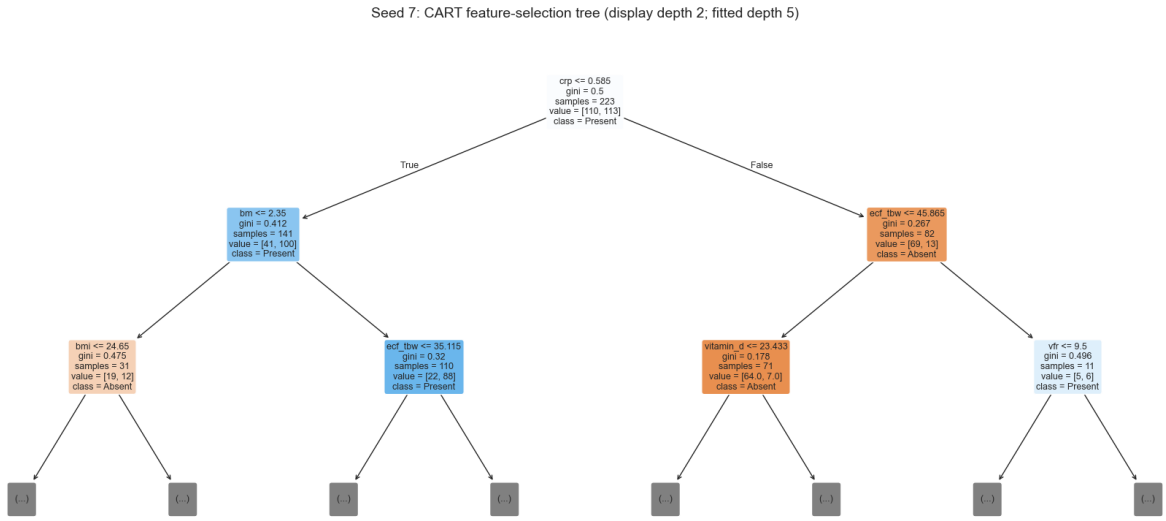


Figure 17: Example CART feature-selection tree for one training split. Only the upper levels of the fitted tree are displayed for interpretability.

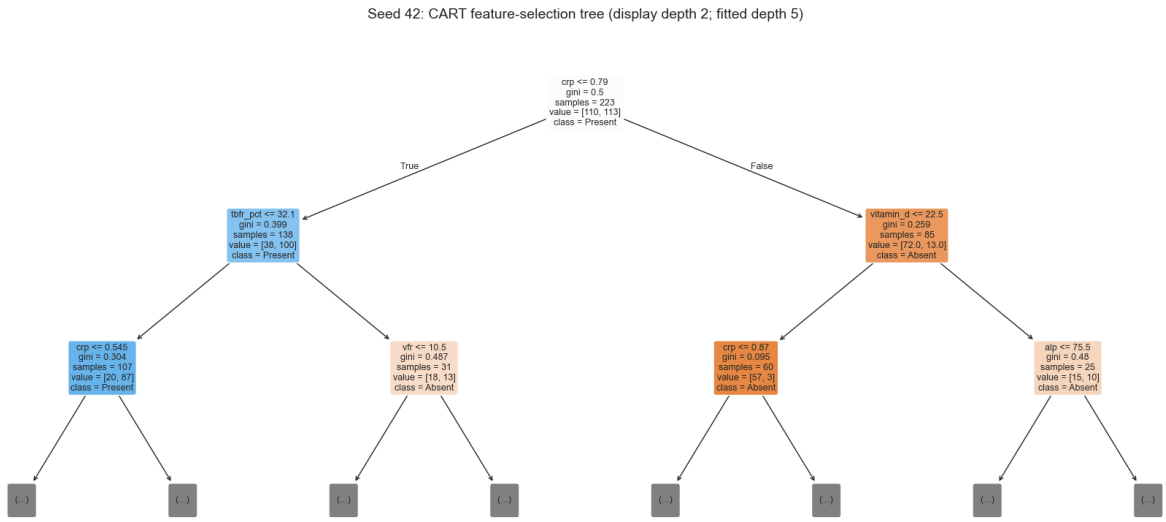


Figure 18: Example CART feature-selection tree from a second training split, illustrating variation in lower-level splits while retaining CRP as the primary separating feature.

The variation in the lower branches across training splits indicates that the dataset does not depend on a single fixed decision rule. Instead, several clinical and body-composition variables provide complementary information. Variables appearing repeatedly across the RFE rankings and CART structures were therefore considered more stable candidates than variables selected only in a single split.

The final number of predictors was not determined using an arbitrary importance threshold. Instead, progressively reduced top-k feature subsets generated by RFE were evaluated on the validation data, and the number of retained

features was selected according to predictive performance across the repeated splits. This provides a direct link between feature reduction and out-of-sample model performance.

Together with the preceding transformation stage, this procedure addresses two forms of data preparation. Statistical transformations project strongly skewed measurements into alternative numerical representations, while RFE and CART provide horizontal reduction by removing predictors that contribute limited additional information. The resulting reduced feature set was subsequently carried forward to model development and compared with the broader predictor set.

Vertical Reduction. No vertical reduction was applied to the dataset. The available sample contains only 319 observations, so removing records without a clear data-quality reason could unnecessarily reduce the amount of information available for training and evaluation. In addition, the target variable is already well balanced, with 161 observations representing gallstones present and 158 representing gallstones absent. The exploratory analysis therefore provided no evidence that undersampling or other record-reduction techniques were required.

All valid observations were consequently retained, while stratified partitioning was used to preserve the target distribution across the training, validation, and test sets. Data reduction in this project instead focused on *horizontal reduction*, where feature-selection methods were used to identify and retain predictors that were most relevant to *Gallstone Status*.

4.7.2 Statistical Transformation

Exploratory analysis identified substantial skewness in several continuous predictors, particularly `obesity_pct`, ALT, AST, HDL, glucose, CRP, triglyceride, GFR, and VFA. Rather than transforming all numerical variables, transformation was considered only for features with $|\text{skewness}| \geq 1$.

Two transformation approaches were evaluated: the logarithmic transformation and the Yeo–Johnson power transformation. Their effectiveness was assessed by comparing the resulting distributions and sample skewness with the original measurements.

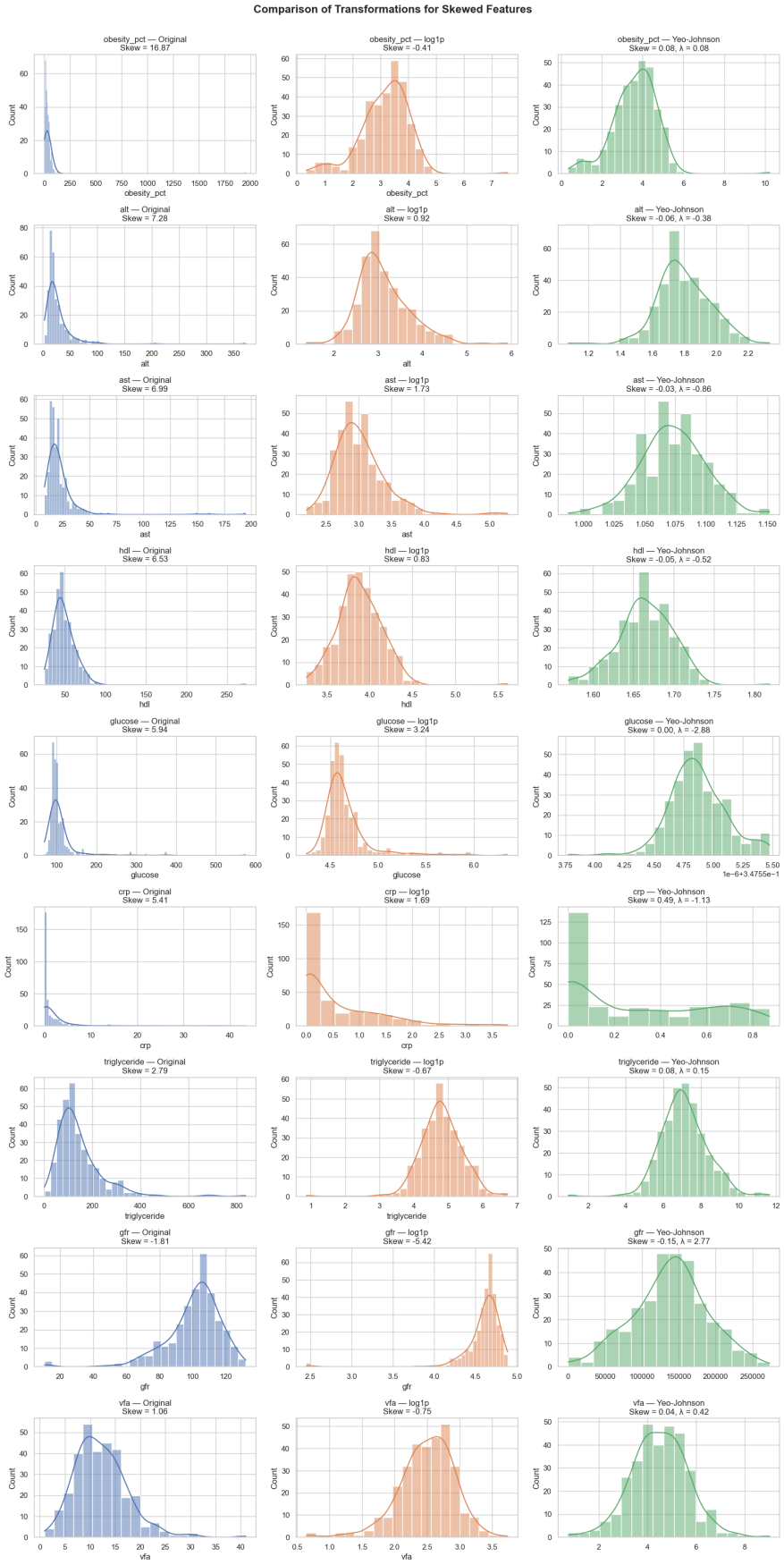


Figure 19: Comparison of the original, `log1p`, and Yeo–Johnson distributions for continuous variables with absolute skewness greater than or equal to 1. Reported values show the resulting skewness and the estimated Yeo–Johnson parameter λ.

Logarithmic Transformation For non-negative variables, a logarithmic transformation was evaluated using

$$x_{\log} = \log(1 + x),$$

where the addition of one allows zero-valued observations to be retained.

As shown in Figure 19, the transformation substantially reduced positive skewness for several variables. For example, the skewness of `obesity_pct` decreased from 16.87 to approximately −0.41, while ALT decreased from 7.28 to 0.92 and triglyceride from 2.79 to 0.67. However, the improvement was not consistent across all predictors. Glucose, for example, remained noticeably right-skewed after transformation, with skewness decreasing from 5.94 to 3.24.

The logarithmic transformation was therefore treated as a feature-specific preprocessing option rather than being applied uniformly to all continuous variables.

Yeo–Johnson Transformation The Yeo–Johnson transformation was also evaluated as a more flexible power transformation. Unlike conventional logarithmic or Box–Cox transformations, the Yeo–Johnson method can accommodate zero and negative values while estimating an appropriate transformation parameter, λ, from the data (Yeo & Johnson, 2000).

The transformation is defined piecewise according to the value of x and the estimated parameter λ, allowing the shape of the transformation to adapt to each variable. This makes it particularly useful when continuous predictors differ substantially in scale and distribution.

Figure 19 shows that Yeo–Johnson generally produced stronger reductions in absolute skewness for the most asymmetric variables. For example, skewness decreased from 16.87 to approximately 0.08 for `obesity_pct`, from 7.28 to −0.06 for ALT, from 6.99 to 0.03 for AST, and from 5.94 to approximately 0.00 for glucose. Triglyceride was similarly reduced from 2.79 to approximately 0.08.

These results indicate that Yeo–Johnson provided a more consistent normalisation of strongly skewed predictors than the logarithmic transformation for this dataset. Nevertheless, improved symmetry alone was not used as sufficient justification for retaining a transformed feature. Transformed and original representations were subsequently evaluated through feature selection and predictive model performance.

Transformation within the Modelling Pipeline Transformations were implemented as part of the modelling pipeline rather than as permanent modifications to the source data. Original measurements were retained for traceability, while transformation parameters were estimated using the training data only and then applied unchanged to the corresponding validation and test partitions.

This procedure prevents information from held-out observations from influencing preprocessing decisions and reduces the risk of data leakage. The final choice between original and transformed representations was therefore based on both distributional improvement and downstream predictive performance rather than on normality alone.

5 Data-Mining Method Selection

5.1 Learning Paradigm and Research Objective

The objective is to estimate gallstone presence from demographic, clinical, laboratory, and body-composition measurements. All 319 records have a known binary outcome, making **supervised learning** the appropriate primary paradigm: predictions can be compared with observed labels on held-out records. **Unsupervised learning** can reveal structure without those labels, but similarity between patients does not necessarily correspond

to disease status. It therefore supports exploration rather than providing the principal answer to the research question.

5.2 Comparison of Data-Mining Methods

The learning paradigm must be narrowed to a method whose output matches the objective. Table 3 distinguishes this decision from the choice of a particular algorithm.

Table 3: Suitability of data-mining methods for the gallstone-prediction objective.

Method	Intended output	Suitability for this project
Classification	A known class and, where supported, its probability.	Directly matches labelled gallstone presence/absence and permits evaluation of discrimination, missed cases, and false alarms. Selected as the primary method.
Continuous-value prediction (regression)	A numerical quantity, such as a laboratory concentration or severity score.	The target is a category, not a continuous measurement. This would answer a different question. Classification is also predictive; the distinction here concerns the type of output.
Segmentation (clustering)	Groups of observations with similar predictor profiles.	Useful for exploratory subgroups, but groups need not correspond to gallstone status. Cluster membership alone is not a validated disease prediction.
Association discovery	Frequently co-occurring conditions or measurement patterns.	Useful for hypothesis generation, but support and confidence do not establish held-out discrimination or an acceptable false-negative rate. Outcome-directed rules would still require supervised classification evaluation.
Anomaly detection	Observations that differ from a reference pattern.	Supports review of unusual measurements, but gallstone cases comprise roughly half the sample and cannot be equated with anomalies.

Dimensionality reduction and feature selection are supporting preparation methods, not substitutes for predicting the labelled outcome. Likewise, forecasting future onset or modelling time to diagnosis would require longitudinal or event-time information not used by the current workflow. The available evidence supports classification of recorded gallstone status, not a demonstrated forecast of future disease.

5.3 Selected Method and Success Criteria

The selected method is **supervised binary classification with probability estimation**. It provides the required patient-level output and directly supports the project benchmarks of ROC–AUC at least 0.80 and gallstone-class sensitivity at least 0.80. Specificity, precision, F1, stability, and interpretability are needed alongside these measures: predicting every patient as positive could satisfy sensitivity while producing unacceptable false alarms.

The source coding is 0 = gallstones present and 1 = absent. The Python modelling target is recoded to 1 = present and 0 = absent, so $\mathrm{FNR} = \mathrm{FN}/(\mathrm{FN} + \mathrm{TP}) = 1 - \text{sensitivity}$ measures missed gallstone cases. These are analytical project benchmarks, not clinical deployment thresholds. The following section compares algorithms within this chosen method and explains how the implemented models are built and selected.

6 Algorithm Selection and Parameterisation

6.1 Range of Classification Algorithms

Having selected supervised classification, the next question is which algorithm offers an appropriate balance of discrimination, case detection, and interpretability for a small tabular dataset. Table 4 distinguishes the algorithms implemented in the Python experiments from alternatives considered on methodological grounds. Unimplemented alternatives are not claimed to have been empirically outperformed.

Table 4: Algorithm-level exploration within supervised classification.

Algorithm	Strength and limitation	Role in this iteration
Logistic regression	Regularised, interpretable probability model; its additive form can miss interactions unless represented in the predictors.	Implemented as the principal baseline. Despite its name, it models a binary class probability here.
Decision trees: CART	Readable nonlinear rules, but individual trees can be unstable and overfit small samples.	CART is used for feature selection;
Random forest	Combines trees to model interactions; less transparent than a single tree, with potentially misleading impurity importance for correlated predictors.	Implemented as the bagged-tree alternative.
XGBoost and LightGBM	Boosted trees can capture nonlinear structure; complexity requires control through tree size, learning rate, and regularisation.	Both implemented with bounded settings and tuning grids.
Support vector machine	Linear or kernel boundaries offer alternatives to additive probabilities; scaling, kernel choice, and probability estimation add decisions.	Considered but not implemented in this bounded Python comparison. Earlier test-set LSVM results are not a selection justification.
k-nearest neighbours and naive Bayes	Neighbour methods are sensitive to scaling and redundant dimensions; naive Bayes is simple but assumes conditional independence.	Not implemented; correlated measurements make their assumptions or distance representation worth further investigation rather than automatic adoption.
Neural network	Can represent complex interactions, but adds architecture and optimisation choices and is harder to interpret.	Deferred given the sample size and scope; no claim that it cannot perform competitively.
Stacking	Combines complementary model probabilities; requires out-of-fold training and increases fitting and explanation costs.	Implemented using the four individual predictive models and a logistic meta-classifier.

This candidate set tests whether nonlinear models or their combination improve on the simpler baseline. A training-prior dummy classifier provides a feature-free benchmark. Unsupervised algorithms are not additional candidates for the final classifier; their exploratory role is established in Section 5.2.

6.2 Model Construction Protocol

The Open Source Software Stack uses Python/Jupyter, pandas and NumPy, scikit-learn, XGBoost, LightGBM, and plotting libraries. Shared `utils.py` and `modeling.py` functions support the transformation, holdout, leave-one-out, tuning, and error-analysis notebooks.

Five label-stratified splits (seeds 42, 7, 21, 84, and 123) each produce 223 training, 48 validation, and 48 test rows. Within training, continuous measurements are median-imputed and categorical fields most-frequent-imputed. RFE with CART or direct CART importance selects 20 source predictors, followed by Yeo–Johnson and standardisation for continuous fields and one-hot encoding for categories. CART selection uses maximum depth 5

and minimum leaf size 4; RFE removes one feature per step. The complete preparation is refitted inside inner CV, including stacking's base pipelines. The globally transformed reference CSV is not an evaluation input.

6.3 Parameter Settings, Rationale and Tested Alternatives

Table 5: Implemented fixed settings and training-only search spaces.

Model	Fixed settings	Tuning grid
Logistic regression	$C = 1$; maximum 3,000 iterations.	$C \in \{0.1, 1, 10\}$.
Random forest	60 trees; maximum depth 5; minimum leaf size 4.	Depth $\{3, 5, 8\}$; leaf size $\{2, 4\}$.
XGBoost	60 trees; learning rate 0.05; depth 2; $\lambda = 1$; row/column sampling 0.8; histogram tree method.	Depth $\{2, 3\}$; learning rate $\{0.03, 0.05\}$.
LightGBM	60 trees; learning rate 0.05; 7 leaves; minimum child samples 20; column sampling 0.8.	Leaves $\{7, 15\}$; learning rate $\{0.03, 0.05\}$.
Stacking	Three stratified inner folds; probability inputs; standardised logistic meta-model, $C = 1$, maximum 3,000 iterations; no raw-feature passthrough.	Fixed benchmark, not tuned.

The tuning notebook fixes RFE–CART and uses three-fold stratified `GridSearchCV` within training to select each family's parameters by mean ROC–AUC. Validation then selects the model family. These limited grids control experiment scope; they do not establish globally optimal settings.

Rationale and limits of the choices. Logistic regression's $C = 1$ provides a regularised reference; the tested $C = 0.1$ and $C = 10$ alternatives increase and decrease regularisation. The 3,000-iteration limit provides optimisation headroom, not additional model capacity. Random forest depth 5 and leaf size 4 limit small-sample fragmentation; depths 3 and 8 and leaf size 2 test that restriction. For XGBoost, depth 2 restricts interaction complexity, learning rate 0.05 shrinks each update, and row/column sampling 0.8 reduces reliance on any one subset. Its depth and learning-rate alternatives test capacity and shrinkage. LightGBM's 7 leaves and minimum child size 20 limit small terminal groups; 15 leaves and learning rate 0.03 test alternatives. The shared 60-tree budget bounds computation; its optimality was not tested. XGBoost's $\lambda = 1$ and histogram method, and the sampling fractions, remain fixed controls rather than empirically optimised choices.

Stacking uses three stratified folds to construct out-of-fold probability inputs within the training data. A standardised logistic meta-model with $C = 1$ and no raw-feature passthrough limits the second-stage input space. The fold count and meta-model settings were not tuned. CART selection's depth 5 and leaf size 4 similarly constrain small-sample splits; RFE removes one field at a time to reach 20. Section 8.5 tests the alternative of 10 fields. The 0.5 threshold is a fixed reference, not a sensitivity-optimised operating point. These distinctions separate tested choices from settings that still require investigation.

Table 6: Executed training-CV alternatives: seed 42, RFE–CART. Winning configurations are selected within each family; scores are not test results.

Model	Trials	Best tested settings	CV AUC
Logistic regression	3	C=0.1	0.8716
Random forest	6	max depth=5, min samples leaf=2	0.8475
XGBoost	4	learning rate=0.05, max depth=3	0.8446
LightGBM	4	learning rate=0.03, num leaves=7	0.8485

Across seeds 42, 7, 21, 84 and 123, the selected logistic C values are 0.1, 0.1, 0.1, 1 and 10. The figure below exposes the actual seed-42 alternatives, rather than only listing a proposed search grid. Source: `results/tuning/seed_42_*_cv.csv`.

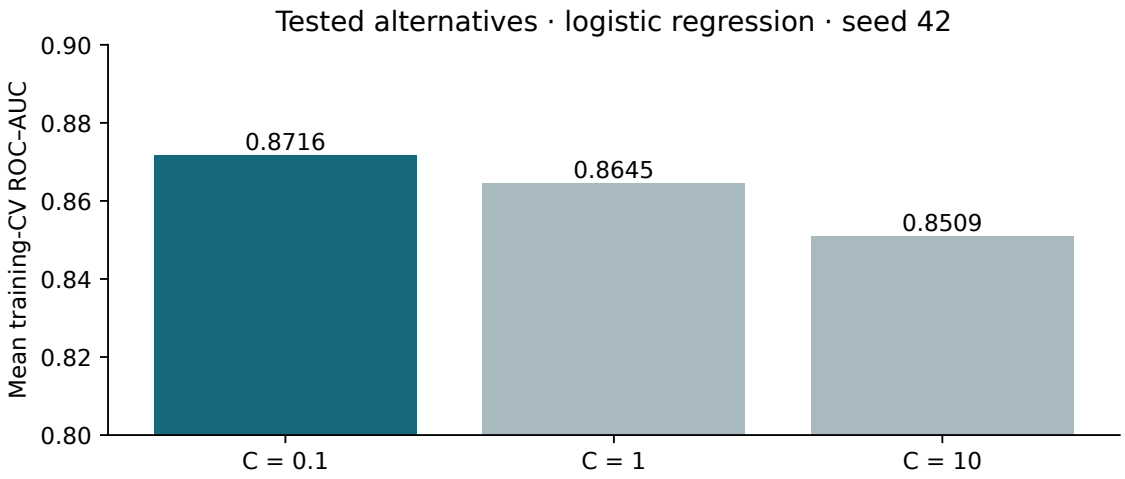


Figure 20: Executed logistic-regression parameter alternatives for seed 42. Scores are means over three training CV folds from the saved search output; validation and test scores are not used in this parameter comparison.

7 Model Execution and Statistical Results

7.1 Execution Protocol and Output Provenance

The construction and settings are specified in Sections 6.2 and 6.3. The evidence below is generated from saved Python result files. It documents completed fits and held-out predictions; it is not an SPSS interface capture or a new model run. The holdout workflow compares five predictive families and a training-prior dummy using both CART and RFE–CART selectors across five seeds. The tuning workflow searches four individual families; stacking remains fixed.

7.2 Evidence of Executed Models

Table 7: Execution inventory counted from saved metrics and prediction files. Counts include the dummy where present and repeated evaluations of the same source records.

Workflow	Metric rows	Prediction rows	Model families
Fixed holdout	120	5,760	6
Tuned holdout	25	1,200	4
Leave-one-out	6	1,914	6

Sources: `results/holdout/`, `results/tuning/` and `results/lopo/`, each containing `metrics.csv` and `predictions.csv`. Prediction rows are evaluation appearances, not unique patients. Saved fitted pipelines are in `models/holdout/`, `models/tuned/` and `models/lopo/`.

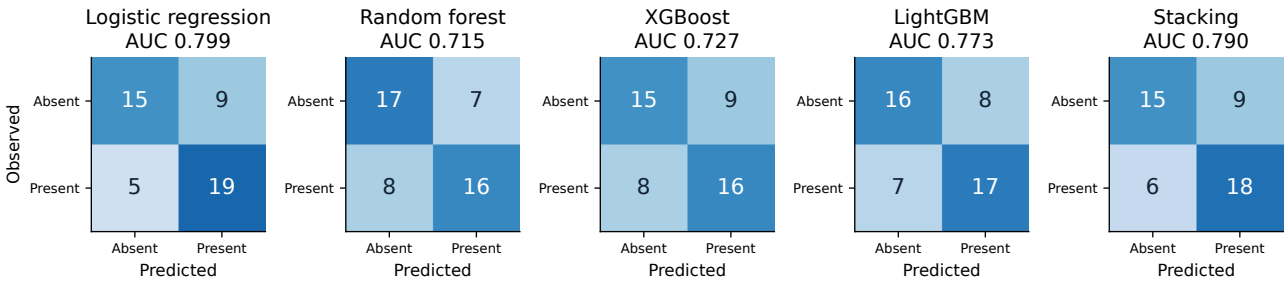


Figure 21: Saved seed-42 validation outputs for all predictive families with RFE–CART. Each confusion matrix contains 48 predictions at threshold 0.5; class 1 denotes gallstones present. These are rendered numerical outputs, not screenshots. Source: `results/holdout/metrics.csv`.

7.3 Statistical Patterns from the Fitted Models

This section records numerical evidence. Interpretation is in Section 8.1, charts in Section 8.2, and evaluation against the project criteria in Section 8.4.

Holdout validation outputs. Table 8 summarises the saved validation results, not test rankings. Each cell reports mean ± sample variance across five holdout seeds, using denominator $5 - 1$. This notation reports variance, not standard deviation or a confidence interval. RFE denotes RFE–CART; bold identifies the highest mean among predictive models in each column, with the dummy shown separately as a benchmark.

Table 8: Five-seed holdout validation: mean ± sample variance at threshold 0.5.

	ROC–AUC		Sensitivity	
Model	CART	RFE	CART	RFE
Logistic regression	**0.834 ± 0.001306**	**0.838 ± 0.001930**	**0.808 ± 0.002257**	**0.758 ± 0.002951**
Random forest	0.798 ± 0.002135	0.776 ± 0.001504	0.717 ± 0.004688	0.717 ± 0.002083
XGBoost	0.791 ± 0.001392	0.777 ± 0.001080	0.742 ± 0.003819	0.708 ± 0.000868
LightGBM	0.791 ± 0.000612	0.772 ± 0.000973	0.717 ± 0.006424	0.717 ± 0.002083
Stacking	0.822 ± 0.000825	0.812 ± 0.000558	0.758 ± 0.005556	0.733 ± 0.000521
Training-prior dummy	0.500 ± 0.000000	0.500 ± 0.000000	1.000 ± 0.000000	1.000 ± 0.000000

Leave-one-out outputs. Table 9 reports the saved LOPO results from `results/lopo/metrics.csv`. Each model has 319 pooled held-out predictions, using RFE–CART selection and seed 42. This is row-level LOOCV, equivalent to leave-one-person-out only if each row is a different person. There is one complete run, so no between-seed variance is reported. Bold marks the highest predictive-model score in each column, except FNR, where lower is better. These are complementary evaluation results, not a new basis for overriding the validation-selected model family.

Table 9: LOPO evaluation on 319 held-out predictions per model at threshold 0.5.

Model	Accuracy	ROC–AUC	Sensitivity	Specificity	F1	FNR
Logistic regression	0.784	**0.874**	0.820	0.747	0.793	0.180
Random forest	**0.799**	0.857	**0.839**	0.759	**0.808**	**0.161**
XGBoost	0.762	0.835	0.807	0.715	0.774	0.193
LightGBM	0.787	0.851	0.832	0.741	0.798	0.168
Stacking	0.787	0.865	0.801	**0.772**	0.791	0.199
Training-prior dummy	0.505	0.000	1.000	0.000	0.671	0.000

Numerical comparison of holdout and LOPO. The comparison fixes the algorithm settings and RFE–CART selector, avoiding confounding evaluation method with CART versus RFE selection. Holdout test entries describe every fixed candidate for comparison; they are not the validation-selected procedure reported in Section 8.4. Sources are `results/holdout/metrics.csv` and `results/lopo/metrics.csv`. Interpretation and differences between the evaluation designs are discussed in Section 8.1.2.

Table 10: Holdout and LOPO comparison for fixed RFE–CART candidates at threshold 0.5. Holdout entries average five seeds; LOPO entries pool 319 held-out predictions from one complete run.

	ROC–AUC			Sensitivity		
Model	Holdout val.	Holdout test	LOPO	Holdout val.	Holdout test	LOPO
Logistic regression	0.838	0.887	0.874	0.758	0.858	0.820
Random forest	0.776	0.850	0.857	0.717	0.850	0.839
XGBoost	0.777	0.844	0.835	0.708	0.858	0.807
LightGBM	0.772	0.841	0.851	0.717	0.850	0.832
Stacking	0.812	0.881	0.865	0.733	0.858	0.801

Conditional associations and extracted rules. Across five 20-feature RFE logistic fits, CRP, vitamin D, ECF/TBW and haemoglobin were always selected, with mean transformed-scale coefficients -1.318, $+0.690$, $+0.810$ and $+0.549$, respectively. Their mean validation permutation AUC decreases were 0.1127, 0.0437, 0.0597 and 0.0172. CRP was the root of all five direct CART selection trees: its recorded-scale cut-point was 0.79 for seed 42 and 0.585 for the other four seeds. The following counts describe seed-42 training nodes, not held-out risks.

Table 11: Conditional profiles extracted from the seed-42 CART selection tree.

Model-defined subgroup	Present / total	Present (%)
CRP ≤ 0.79	100/138	72.5
CRP > 0.79	13/85	15.3
CRP ≤ 0.79, total-body-fat ratio ≤ 32.1	87/107	81.3
CRP ≤ 0.79, total-body-fat ratio > 32.1	13/31	41.9
CRP > 0.79, vitamin D ≤ 22.5	3/60	5.0
CRP > 0.79, vitamin D > 22.5	10/25	40.0

8 Interpretation, Evaluation and CRISP-DM Iteration

8.1 Discussion of the Statistical Patterns

Logistic regression has the strongest mean validation AUC under both selectors. CART selection gives it higher sensitivity and F1 (0.761 versus 0.749 for RFE), despite slightly lower AUC. Stacking has lower between-seed AUC variances (0.000825 with CART and 0.000558 with RFE) than logistic regression (0.001306 and 0.001930), but also lower discrimination and sensitivity. Thus, stability alone does not justify the added complexity. The dummy's perfect sensitivity accompanies zero specificity and accuracy 0.500, demonstrating why sensitivity cannot be the only criterion.

The implemented ranking uses descending validation AUC, descending accuracy, and ascending FNR, followed by deterministic name-based tie-breaks. F1, stability, and interpretability inform the discussion but are not numerical components of this ranking. Logistic regression wins all five fixed-model comparisons: CART is selected for seeds 42, 21, and 123, and RFE for seeds 7 and 84. This supports the simpler model family, not a universally best selector or a claim of statistically significant superiority.

Tuning also selects logistic regression in all five seeds. The selected C values are 0.1, 0.1, 0.1, 1, and 10 in the

stated seed order. These choices come from training CV and validation, not test performance. Selection does not guarantee the success criteria: for example, the seed-7 fixed-model winner has validation sensitivity 0.708. Recall is not a hard eligibility constraint, and threshold optimisation and dedicated calibration assessment remain unimplemented.

8.1.1 Interpretation of the Leave-One-Out Results

The LOPO results in Table 9 show a trade-off between ranking patients by predicted probability and classifying them at threshold 0.5. Logistic regression has the highest ROC–AUC (0.874), so it provides the strongest overall discrimination among the tested models. Its sensitivity of 0.820 corresponds to identifying 132 of 161 gallstone records and missing 29; specificity of 0.747 corresponds to correctly classifying 118 of 158 negative records, with 40 false positives. Thus, meeting both 0.80 project benchmarks still leaves approximately 18% of positive records undetected. AUC alone does not establish calibration or an acceptable operating threshold.

Random forest performs best at the fixed threshold: accuracy is 0.799, sensitivity 0.839 and F1 0.808. It detects 135 positive records, misses 26 and produces 38 false positives. Compared with logistic regression, these aggregate counts represent three fewer missed cases and two fewer false alarms, despite a lower AUC of 0.857. This makes random forest a useful candidate for further sensitivity-oriented validation, but the small observed differences do not establish statistically significant superiority or justify replacing the validation-selected family after inspecting LOPO results.

Stacking achieves the highest specificity (0.772), with 36 false positives, but misses 32 positive records: its sensitivity of 0.801 only narrowly meets the benchmark. Its extra complexity therefore does not deliver the best case detection. LightGBM offers sensitivity 0.832 with 27 false negatives and 41 false positives, while XGBoost has the lowest AUC (0.835) and accuracy (0.762), with 31 false negatives and 45 false positives. All five predictive models meet the two numerical benchmarks in this pooled run, but their error trade-offs differ. The dummy's sensitivity of 1.000 is uninformative without specificity: it incorrectly flags all 158 negative records. Its unusual pooled AUC is explained below.

8.1.2 Comparison of Holdout and Leave-One-Out Evaluation

Table 12: Roles and limitations of the two implemented evaluation methods.

Dimension	Repeated stratified holdout	Leave-one-out (LOPO)
Training and assessment	223 training, 48 validation and 48 test records per seed. Selected configurations and test-stage candidate fits use the 271 development records.	318 training records and one held-out record per fold; 319 folds cover the whole dataset once for each fixed candidate.
Role in this project	Validation selects the model/selector; training CV selects tuning parameters. Test results assess fitted candidates or the selected procedure.	Provides complementary internal assessment of fixed RFE–CART candidates; it does not repeat the entire model-selection process inside each fold.
Coverage and variation	Five seeds expose split-to-split variation. Validation/test sets contain 48 records each, and records can recur across overlapping splits.	Each source row contributes one held-out prediction per model. There is one complete run; heavily overlapping training folds are not independent repeats.
Computation	Fewer outer fits per fixed candidate; tuning and stacking add inner fits.	319 outer fits per candidate, with additional inner fitting for stacking.
Interpretation boundary	Performance depends on split composition and development size; inspected test partitions are not a sealed external assessment.	Uses more training data per prediction, but offers no external validation and no between-seed variance estimate from the single run.

Table 10 compares the same fixed RFE–CART candidates. Logistic regression's AUC is 0.838 on holdout validation and 0.874 under LOPO, while sensitivity rises from 0.758 to 0.820. Across all five families, LOPO AUC exceeds mean holdout-validation AUC by approximately 0.036–0.081, and sensitivity by 0.062–0.122. The pattern is consistent with differences in training size (318 versus 223 records) and assessment coverage, but this comparison cannot isolate their effects. It is not evidence that changing the evaluation method improved an otherwise identical fitted model.

The holdout *test* comparison gives a different picture. Fixed RFE logistic regression averages AUC 0.887 and sensitivity 0.858 on test, compared with LOPO's 0.874 and 0.820. LOPO AUC is higher than mean holdout test AUC for random forest and LightGBM, but lower for logistic regression, XGBoost and stacking. LOPO sensitivity is lower for all five families. Consequently, LOPO is not uniformly more favourable. Test fits use 271 development records and assess different, overlapping 48-row subsets; LOPO evaluates every row with a separate 318-row training set. These estimates answer related but different questions and should not be ranked as if they were a controlled model-improvement experiment.

The two methods nevertheless agree that logistic regression has the highest AUC among fixed RFE candidates. Their threshold-dependent rankings differ: random forest leads LOPO sensitivity and F1, whereas logistic regression, XGBoost and stacking share the highest mean RFE holdout-test sensitivity when rounded to three decimals. This supports keeping discrimination, sensitivity and false alarms separate in the decision. Use holdout validation for the stated selection procedure and LOPO as a complementary robustness check; neither establishes performance on a new cohort. LOPO is leave-one-person-out only if each source row represents a unique person. The single run supplies no evidence that LOPO has lower uncertainty than repeated holdout, and its folds must not be treated as 319 independent experiments.

Patterns across model families. The tables in Section 7.3 document the outputs for every fixed model. Their patterns, rather than only the leading score, are summarised in Table 13. Differences between holdout validation and LOPO reflect different training sizes and evaluation samples; they do not by themselves establish a statistically significant improvement.

Table 13: Observed performance patterns for every fixed model and benchmark.

Model	Pattern and interpretation
Logistic regression	Leads validation AUC with either selector and retains the highest LOPO AUC. CART improves validation sensitivity and F1 relative to RFE. Strong ranking performance does not eliminate missed cases at threshold 0.5.
Random forest	CART improves validation AUC over RFE without changing mean sensitivity. Under LOPO it has the highest accuracy, sensitivity, and F1, and lowest FNR among predictive models, but lower AUC than logistic regression. Its advantage is threshold-specific case detection rather than the strongest ranking.
XGBoost	CART improves validation AUC and sensitivity over RFE. It has the lowest LOPO accuracy and AUC among predictive models, so the tested boosting configuration provides no consistent advantage over simpler alternatives.
LightGBM	CART improves validation AUC while mean sensitivity is unchanged. LOPO sensitivity exceeds logistic regression's, but AUC and specificity are lower. It offers a different error balance rather than dominance across metrics.
Stacking	Validation AUC varies less across seeds than logistic regression's but has a lower mean. LOPO gives the highest specificity and the highest FNR among predictive models: fewer false alarms accompany more missed positive cases. Combining models does not automatically improve all outcomes.
Training-prior dummy	Predicts every evaluated row as positive at threshold 0.5. Perfect sensitivity and zero FNR therefore accompany zero specificity and near-half accuracy; this is a benchmark failure pattern, not useful disease discrimination.

The tuning experiment also built all four individual model families, not only the selected logistic models. Their

mean validation AUCs were 0.827 for logistic regression, 0.779 for random forest, 0.777 for XGBoost, and 0.767 for LightGBM, from `results/tuning/metrics.csv`. Relative to the fixed RFE comparison, tuning did not consistently improve external-to-search validation AUC: logistic regression and LightGBM decreased, random forest increased slightly, and XGBoost was essentially unchanged at three decimal places. Training-CV optimisation therefore does not guarantee validation improvement. Stacking and the dummy were not tuned.

The dummy's pooled LOPO AUC is zero because leaving out a positive case lowers the training prior while leaving out a negative raises it, reversing the ranking of the fold-specific probabilities. A fixed chance score instead has AUC 0.5.

Interpretation of fitted-model associations. The following evidence comes from fitted training models and validation perturbations, rather than repeated univariate EDA. Continuous logistic coefficients refer to one training-standard-deviation increase *after* Yeo–Johnson transformation. They describe conditional model associations, not causal effects or original-unit clinical changes. CART rules come from the feature-selection tree, not a surrogate explanation of every ensemble.

Recurring signals and their direction. Four predictors recurred in all five 20-feature RFE fits: CRP, vitamin D, the ratio of extracellular fluid to total body water (`ecf_tbw`), and haemoglobin. Their mean logistic coefficients were respectively -1.318, $+0.690$, $+0.810$ and $+0.549$, with the same sign in every split. CRP ranged from -1.513 to -1.154 and vitamin D from $+0.569$ to $+0.801$. Thus the fitted models combine laboratory and body-composition information rather than demographic risk alone.

Each raw predictor was also independently shuffled 20 times on each validation partition. Mean AUC decreases across five fits were 0.1127 for CRP, 0.0597 for `ecf_tbw`, 0.0437 for vitamin D and 0.0172 for haemoglobin. These quantify perturbation-based contributions, not additive percentages of explained disease risk. The prominence of CRP, vitamin D and bioimpedance measurements agrees with the feature families reported by İ. Esen et al. (2024); that study concerns the originating dataset and is not independent replication.

Thresholds and conditional interactions. CRP was the root split in all five direct CART selection trees, but its threshold varied: 0.79 for seed 42 and 0.585 for the other four seeds. Table 11 traces the seed-42 tree before transformation, retaining the recorded dataset scale. Counts describe training nodes, not externally validated risks or calibrated probabilities.

Body-fat ratio separates records within the low-CRP branch, whereas vitamin D separates those within the high-CRP branch. Within the low-CRP, higher-body-fat branch, visceral-fat rating ≤ 10.5 identifies 3/18 positive records, compared with 10/13 above 10.5. This suggests a candidate interaction between overall and visceral adiposity, not an established biological interaction. Different body-composition measures may capture different patient profiles, but these small, adaptively chosen groups require independent confirmation.

Crucially, lower CRP and higher vitamin D correspond to higher predicted gallstone presence here. These directions must not be rewritten as conventional causal risk statements. The outcome is existing gallstones; sampling, treatment or measurement context could affect associations. Learned cut-points are not clinical reference limits. Units and clinical context must be verified before treating any value as a diagnostic threshold; neither increasing CRP nor reducing vitamin D is a supported intervention.

Subgroup differences beyond a global coefficient. The female-versus-male logistic contrast was positive in all five models: the mean difference between their one-hot coefficients was $+0.814$, an odds multiplier of approximately 2.26 for that mean contrast, conditional on other transformed predictors. Nevertheless, seed 42's validation model missed 4 of 14 positive female records and 1 of 10 positive male records (27 female and 21 male records overall). A positive demographic coefficient therefore does not ensure equal identification: other covariates can outweigh it. Small denominators support a targeted audit, not a population-level fairness conclusion.

Hepatic-fat grades did not form a monotonic model pattern. Relative to grade 0, mean logistic contrasts were approximately +0.819 for grade 1, −0.535 for grade 2 and −0.460 for grade 3. These models therefore do not establish that increasing hepatic-fat severity increases gallstone risk. Conditional associations, sparse categories and overlapping metabolic measurements are competing explanations requiring clinical review.

Why weaker predictors contribute less. Selection alone does not establish useful prediction. Alkaline phosphatase was retained in all five RFE fits, yet its mean coefficient was only −0.081, changed sign across fits, and its mean validation permutation contribution was −0.0024. Triglycerides appeared in three fits, had mean coefficient +0.013 when selected, and mean perturbation contribution −0.0055 across all five fits. Negative contributions mean shuffling slightly improved validation discrimination, consistent with unstable or unnecessary fitted information; they do not indicate a protective clinical effect.

An excluded variable necessarily has zero permutation contribution in that pipeline. Age was excluded in every reference fit: this demonstrates non-use by the selector, not clinical irrelevance. Correlated measurements may substitute for each other, and single-variable permutation can understate shared information. RFE also ranks usefulness to CART, which need not match usefulness to the downstream logistic model.

The strongest signal is concentrated in CRP, but complementary contributions remain in fluid balance, vitamin D and smaller laboratory or categorical effects. The later 10-feature comparison in Section 8.5 tests retention of information in aggregate; it does not prove that every retained variable is necessary or identify which removed variable caused the loss. Grouped ablation would be needed to distinguish shared body-composition information from unique contributions.

8.2 Visualisation of Model Patterns

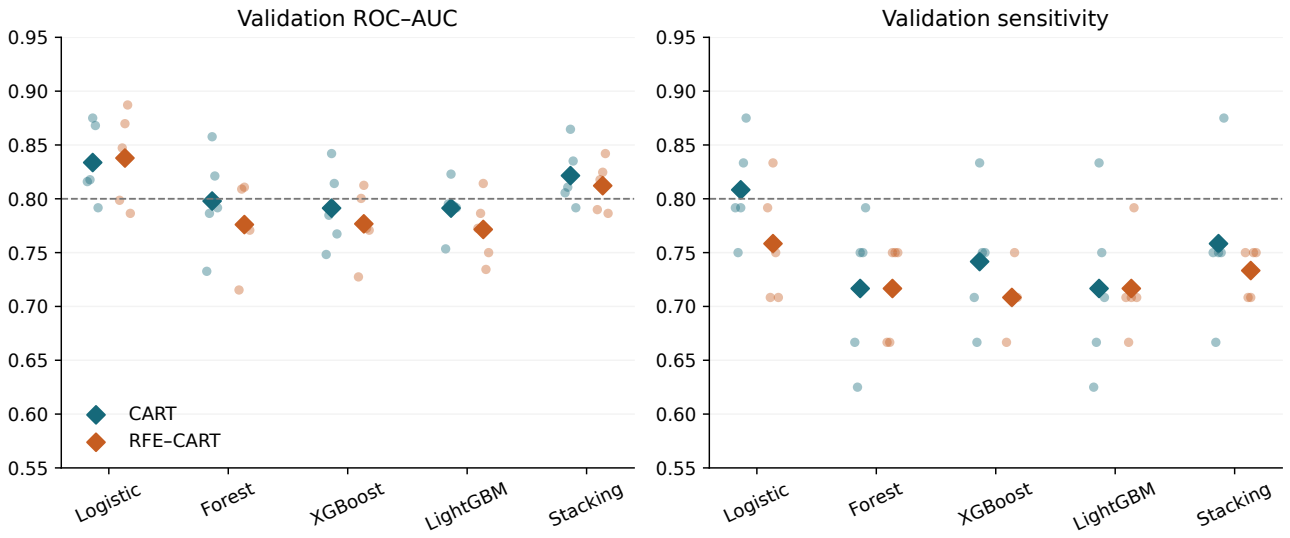


Figure 22: Validation discrimination and sensitivity for every predictive family and selector. Large markers show five-seed means; small markers show individual seeds, not confidence intervals. Dashed lines indicate the project benchmarks. Source: `results/holdout/metrics.csv`.

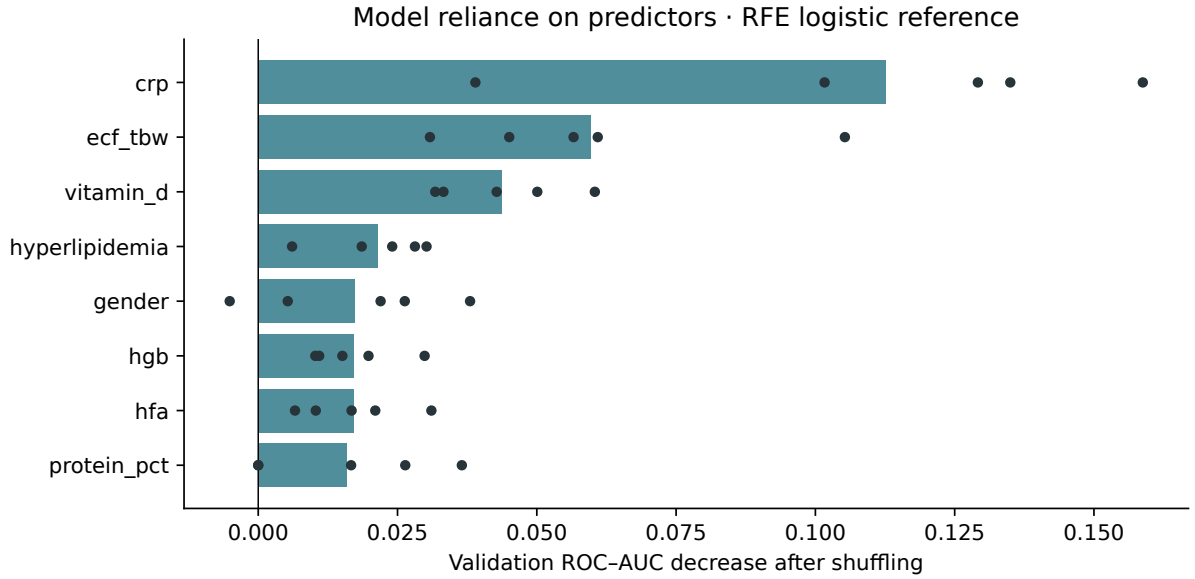


Figure 23: Largest mean validation permutation contributions in the 20-feature RFE logistic reference. Bars are mean AUC decreases; points are five fitted splits, each averaging 20 shuffles. Correlated fields can share information; these values are neither causal effects nor additive risk percentages. Source: `results/iteration_patterns/permutation.csv`.

8.3 Error Analysis: Repeatedly Misclassified Records

The saved prediction files permit review of errors that recur across seeds and model families. Table 14 uses the *validation-selected fixed model* for each holdout seed, identified by `results/holdout/validation_winners.csv`, and its test predictions in `results/holdout/predictions.csv`. All these selected models are logistic regression. Probabilities refer to gallstones being present; predictions use threshold 0.5. The examples were chosen retrospectively to illustrate errors, not to estimate their prevalence in future patients.

Table 14: Examples of repeated held-out errors; denominators count eligible test appearances.

Source row	Holdout seeds	Error	Errors/runs	Probability range	LOPO errors/models
117	7, 42	FN	2/2	0.068–0.350	4/5
291	21, 42, 84	FP	3/3	0.703–0.758	5/5
316	7, 84, 123	FP	3/3	0.634–0.794	5/5

Record 117 has gallstones but was predicted absent in both holdout test appearances. In LOPO, logistic regression, random forest, XGBoost, and stacking also missed it; LightGBM correctly classified it with probability 0.566. The tuned holdout winners still missed it in both seeds. This is a recurring false-negative example, although the LightGBM result shows that the error is not universal across models.

Records 291 and 316 have no gallstones but were repeatedly predicted positive. Across all five predictive families and both selectors, record 291 was wrong in 29/30 holdout predictions and record 316 in 30/30; both were false positives for all five LOPO models. Tuning corrected record 291 only in seed 42, with probability 0.491, just below the decision threshold, whereas record 316 remained wrong in all three tuned test appearances. The first correction is therefore a borderline classification change, not evidence that tuning consistently resolves the record's errors.

These identifiers are zero-based `source_row` values, not verified patient IDs. Repetition means the same dataset record was tested in several overlapping splits or by different models, not that a patient had repeated clinical visits. The LOPO denominator excludes the dummy benchmark. Shared errors across model families motivate

review of source measurements, labels, and overlapping predictor profiles, but do not prove mislabelling or a causal explanation. Their review must not be used to retune models and then claim independent performance on the same test records.

8.4 Model Evaluation Against the Project Objectives

Table 15: Internal test performance of the validation-selected fixed procedure across five seeds. Ranges describe split variation, not confidence intervals.

Metric	Mean	Minimum–maximum
ROC–AUC	0.883	0.828–0.915
Sensitivity	0.892	0.792–0.958
Specificity	0.708	0.667–0.792
F1	0.816	0.769–0.846

These results come from `results/holdout/selected_test_metrics.csv`; selectors vary by seed. They describe the selected procedure, whereas the all-candidate tables in Section 7.3 describe individual models. Reused, overlapping test partitions limit independence. The selected family is logistic regression, chosen using validation rather than test ranking. Mean scores can meet a benchmark while individual splits fail it. At the fixed 0.5 threshold, the RFE logistic reference has mean validation AUC 0.838 but sensitivity 0.758: discrimination meets 0.80, whereas case detection does not. The CART logistic candidate has means 0.834 and 0.808, but this does not establish reliable performance for every split. LOPO logistic AUC 0.874 and sensitivity 0.820 are complementary internal results for a fixed candidate, not a nested evaluation of selection. The 20-feature RFE reference is consequently used for a controlled follow-up experiment in Section 8.5; it is not claimed to be the selected selector in every seed.

Final fitting and evaluation integrity. Selected configurations are refitted on the 271 development rows before assessment on the 48 test rows. Complete pickle pipelines preserve preparation and classifier state in `models/holdout/` and `models/tuned/`. LOOCV evaluates fixed candidate pipelines by holding out each row once; `models/lopo/` contains separate full-data refits for future inference. It is row-level evaluation, and it is not nested evaluation of the entire algorithm-selection procedure.

The Iteration 2 concern is **partly resolved**: automatic selection now excludes test scores, and the model-building argument above uses validation evidence. However, the fixed-model notebook still evaluates every candidate on test before recording its validation winner, and tuning reuses those partitions. The resulting exposure permits informal test-driven decisions. Overlapping splits and prior dataset exploration further limit independence; the results must be described as internal evaluations.

A stricter future workflow would lock the complete configuration before accessing an untouched test set and evaluate only that configuration. Alternatively, nested outer evaluation should contain all selection decisions within its training folds. Error analysis may motivate later revisions, but revised models require new assessment. This qualification addresses the feedback without claiming that the current implementation provides a fully sealed final test.

Decision and feedback trigger. Retain logistic regression as the leading research family, but do not claim that the current workflow establishes deployment readiness. Calibration, threshold selection, generalisation and the cost of false positives remain unresolved. Variable sensitivity and recurring errors motivate returning to feature selection after this evaluation. The next subsection records the completed controlled revision, including its unsuccessful outcome.

8.5 A Completed CRISP-DM Feedback Cycle

The Python/Jupyter workflow follows CRISP-DM as a feedback cycle. The business objective remains to support assessment of gallstone presence, with missed cases a particular concern; this is not a prospective disease-onset model. Sections 7 and 8.4 report execution evidence and the model evaluation. Here, evaluation motivates a return to feature selection, followed by an executed rerun and a documented decision.

Cycle A: establish and evaluate the reference. Data understanding established the coded categories and outcome direction: source status 0 becomes modelling class 1 (gallstones present). Data preparation fitted imputation, CART-based recursive feature elimination (RFE; 20 predictors), Yeo–Johnson transformation and standardisation on each training partition. Logistic regression used $C = 1$ and a 0.5 classification threshold. Five stratified 70–15–15 splits used seeds 42, 7, 21, 84 and 123, with 223 training, 48 validation and 48 test records each. Evaluation exposed variable sensitivity and recurring errors, prompting the question: does a smaller predictor set remove unstable information without losing useful signal?

Cycle B: revisit preparation, rebuild and re-evaluate. A focused revision reduced RFE from 20 to 10 predictors. This was the only changed modelling choice: rows, seeds, transformations, logistic settings and threshold remained fixed. All preprocessing and feature selection were fitted again on the 223 training records, and both cycles were rerun on the corresponding validation records. No test records were evaluated in this additional experiment. This completes the loop *evaluation* → *revised data preparation* → *modelling* → *evaluation* → *decision*, rather than merely proposing future work.

Table 16: Paired validation comparison. Summary entries are mean ± sample variance across five seeds, not standard deviation. Bold indicates the higher mean or seed-level value.

	Validation AUC		Sensitivity	
Seed	20 features	10 features	20 features	10 features
42	**0.799**	0.760	**0.792**	**0.792**
7	**0.847**	0.825	**0.708**	0.667
21	**0.870**	0.826	**0.833**	0.750
84	**0.887**	0.856	**0.750**	**0.750**
123	0.786	**0.811**	0.708	**0.792**
Mean ± variance	**0.838** ± 0.001930	0.816 ± 0.001223	**0.758** ± 0.002951	0.750 ± 0.002604

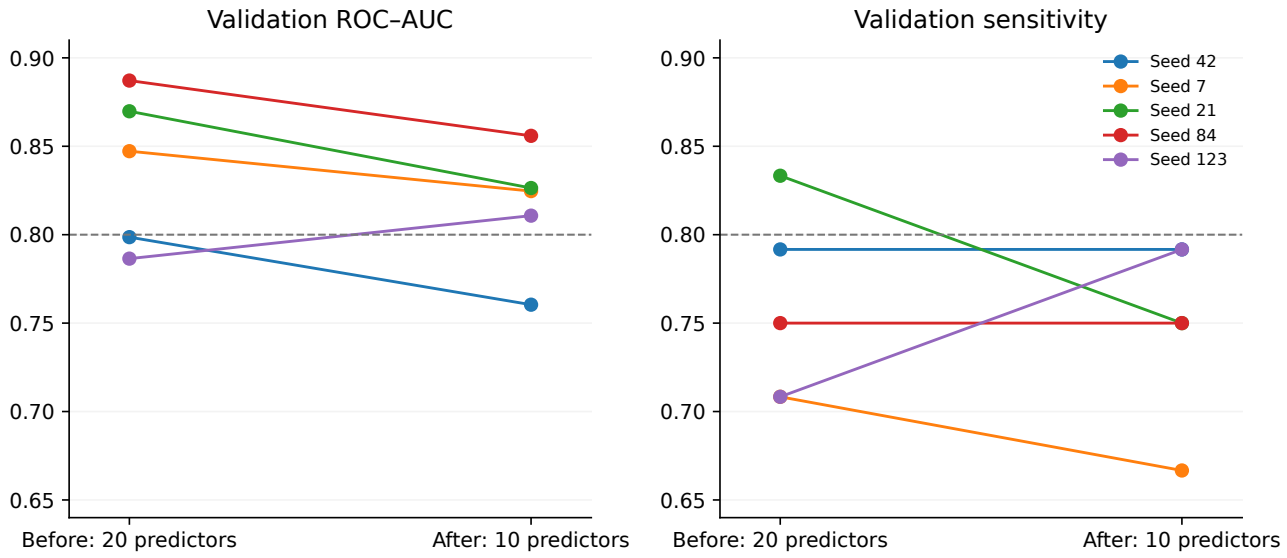


Figure 24: Completed feedback cycle: paired validation results before and after reducing RFE from 20 to 10 predictors. Each line connects the same seed; all other modelling choices are held fixed. Dashed lines mark the 0.80 project targets. Source: `results/iteration_patterns/cycle_metrics.csv`.

Decision and robustness boundary. The revision lowered AUC in four of five splits, with a mean change of −0.0222; sensitivity fell by 0.0083 on average. Lower between-seed variance did not compensate for lower discrimination or resolve missed cases. The 20-feature reference was therefore retained for this comparison. This tests robustness to a specific simplification, not universal superiority: validation partitions overlap across seeds, have already informed development, and neither cycle achieved mean sensitivity of 0.80. The revised model was not promoted to the saved candidate folders. Separate tuning and LOPO results remain in Sections 6.3 and 7.3; they are not independent external confirmation of this iteration.

Proposed next feedback cycle. The completed comparison shows that reducing the number of predictors alone does not resolve missed cases. Table 17 translates this finding and the remaining evaluation gaps into specific preparation and modelling changes. These are proposed experiments, not additional completed cycles. Each experiment should record its motivating finding, controlled change, evaluation result and retain-or-reject decision.

All feature, model, calibration and threshold choices must remain inside the development process. Because the current test partitions have already been inspected, subsequent results on those records remain exploratory. Nested evaluation can assess a prespecified selection procedure internally, while an untouched external cohort is needed to examine transfer beyond this dataset. If no candidate satisfies the predefined success criteria, the next cycle should report that outcome rather than select a model by AUC alone.

8.6 Return to Business Understanding and Deployment

The completed cycle supports a research screening aid, not deployment or replacement of clinical assessment. Simplification can sacrifice complementary information without resolving false negatives. Repeatedly misclassified records documented in Section 8.3 should guide source-data and subgroup review, not retrospective relabelling. The next CRISP-DM cycle should specify calibration and sensitivity-oriented threshold selection within development data, then assess the frozen pipeline on an untouched external cohort. These steps are planned, not claimed as completed.

Table 17: Proposed changes for the next CRISP-DM feedback cycle.

Area	Proposed change and rationale	Evaluation and decision rule
Feature engineering	Test a small, prespecified set of representations or interactions, such as overall-by-visceral adiposity, motivated by conditional model patterns. Verify units and measurement availability; distinguish existing ratios from newly engineered features.	Compare against the unchanged logistic reference using paired development folds. Fit transformations within training folds and retain changes only when validation supports their contribution.
Feature importance	Use grouped permutation and refitted group ablation for correlated fluid, adiposity and laboratory measurements, extending individual importance estimates.	Compare AUC, sensitivity and stability. Distinguish a fitted model's reliance on a variable group from the performance achievable after refitting without it; low individual importance alone does not justify removal.
Feature selection	Compare the 20-feature RFE reference with logistic-compatible selection, regularised logistic regression using all eligible predictors, and a predefined routine-measurement subset.	Choose feature counts and penalties within inner development folds. Assess performance and measurement burden together; the unsuccessful 10-feature revision does not establish that every retained field is necessary.
Calibration and threshold	Assess probability calibration and select a threshold using a prespecified sensitivity target and an agreed specificity or assessment-workload constraint.	Fit any calibrator and choose the threshold within development data using held-out predictions. Freeze both before outer evaluation; report calibration, sensitivity, specificity and confusion counts.
Subgroup evaluation	Extend the preliminary sex-specific error review to predefined sex and age groups, checking measurement quality and sample support.	Report group sizes, positive counts, error rates and uncertainty. Small groups support further investigation rather than firm fairness claims.

Reproducibility. The executed comparison and pattern extraction are implemented in `scripts/iteration_pattern_analysis.py`, using the existing `utils.py`. The folder `results/iteration_patterns/` records per-seed cycle results, sample variances, selected fields, coefficients, validation permutation contributions, subgroup counts, CART node counts and the source-data signature. The analysis refits training-specific transformations from raw records; it does not evaluate the full-data reference transformed CSV.

9 Acknoledgement

I acknowledge that this submission is my own original work and has been prepared in accordance with the University of Auckland's policies on academic integrity and copyright. Further information is available from the University of Auckland Academic Integrity and Copyright guidelines.

I also confirm that I have the appropriate permission to use the data included in this project. Where data are owned by an organisation and are not publicly available, permission must be obtained from the relevant rights holder, including permission to upload the data to Canvas. The University of Auckland accepts no responsibility for any misuse of data by the student.

Use of Artificial Intelligence

Generative artificial intelligence tools were used to support language editing, sentence restructuring, LaTeX formatting, and the organisation of the report. All project decisions, data analysis, interpretation of results, and final content were reviewed and completed by the author. No confidential, personal, or restricted data were shared with an artificial intelligence tool.

References

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Esen, I., Arslan, H., Aktürk, S., Gülşen, M., Kültekin, N., & Özdemir, O. (2025). Gallstone [Dataset donated 19 April 2025; based on the clinical study reported by Esen et al. (2024)].

Esen, İ., Arslan, H., Aktürk Esen, S., Gülşen, M., Kültekin, N., & Özdemir, O. (2024). Early prediction of gallstone disease with a machine learning-based method from bioimpedance and laboratory data. *Medicine*, *103*(8), e37258. https://doi.org/10.1097/MD.0000000000037258

European Association for the Study of the Liver (EASL). (2016). EASL clinical practice guidelines on the prevention, diagnosis and treatment of gallstones. *Journal of Hepatology*, *65*(1), 146–181. https://doi.org/10.1016/j.jhep.2016.03.005

Fujita, N., Yasuda, I., Endo, I., Isayama, H., Iwashita, T., Ueki, T., Uemura, K., Umezawa, A., Katanuma, A., Katayose, Y., Suzuki, Y., Shoda, J., Tsuyuguchi, T., Wakai, T., Inui, K., Unno, M., Takeyama, Y., Itoi, T., Koike, K., & Mochida, S. (2023). Evidence-based clinical practice guidelines for cholelithiasis 2021. *Journal of Gastroenterology*, *58*(9), 801–833. https://doi.org/10.1007/s00535-023-02014-6

Gutt, C., Schläfer, S., & Lammert, F. (2020). The treatment of gallstone disease. *Deutsches Ärzteblatt International*, *117*(9), 148–158. https://doi.org/10.3238/arztebl.2020.0148

Lammert, F., Gurusamy, K., Ko, C. W., Miquel, J.-F., Méndez-Sánchez, N., Portincasa, P., van Erpecum, K. J., van Laarhoven, C. J. H. M., & Wang, D. Q.-H. (2016). Gallstones. *Nature Reviews Disease Primers*, *2*, 16024. https://doi.org/10.1038/nrdp.2016.24

National Institute for Health and Care Excellence. (2014). *Gallstone disease: Diagnosis and management* (tech. rep. No. CG188) (Clinical guideline). NICE.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, *12*, 2825–2830.

Salem, N. M., Jack, K. M., Gu, H., Kumar, A., Garcia, M., Yang, P., & Dinu, V. (2023). Machine and deep learning identified metabolites and clinical features associated with gallstone disease. *Computer Methods and Programs in Biomedicine Update*, *3*, 100106. https://doi.org/10.1016/j.cmpbup.2023.100106

van der Maaten, L., & Hinton, G. (2008). Visualizing data using t-sne. *Journal of Machine Learning Research*, *9*, 2579–2605.

Wang, X., Yu, W., Jiang, G., Li, H., Li, S., Xie, L., Bai, X., Cui, P., Chen, Q., Lou, Y., Zou, L., Zhou, Z., Zhang, C., Sun, P., & Mao, M. (2024). Global epidemiology of gallstones in the 21st century: A systematic review and meta-analysis. *Clinical Gastroenterology and Hepatology*, *22*(8), 1586–1595. https://doi.org/10.1016/j.cgh.2024.01.051

Yeo, I.-K., & Johnson, R. A. (2000). A new family of power transformations to improve normality or symmetry. *Biometrika*, *87*(4), 954–959. https://doi.org/10.1093/biomet/87.4.954

A Instructions for Running the Submitted Workflow

This appendix explains how to run the submitted Iteration 3 Python/Jupyter workflow. All commands below are run from the `Iteration3_OpenStack` directory unless stated otherwise. Keep the supplied directory structure intact.

A.1 Environment and Required Files

Install Python and create a dedicated virtual environment. The submitted notebooks record Python 3.13.5; package requirements are minimum versions rather than an exact environment lock. On macOS or Linux, run:

```
cd INFOSYS722/Iteration3_OpenStack
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python -m ipykernel install --sys-prefix \
  --name gallstone-iteration3 \
  --display-name "Gallstone Iteration 3"
```

On Windows PowerShell, create the environment with `python -m venv .venv` and activate it using `.venv\Scripts\Activate.ps1`. Run the kernel-install command on one line, omitting the shell continuation backslashes.

The active modelling workflow requires `data/raw/dataset_gallstone`, which is a CSV file with no extension. Preserve its filename and row order. The accompanying `data/raw/dataset_gallstone.xlsx` is also supplied, but does not replace the extensionless CSV used by the modelling functions. Keep `utils.py`, `modeling.py`, `requirements.txt`, `run_workflow.py`, and the `notebooks/`, `scripts/`, `tests/` and `Report/` directories together with the raw data.

A.2 Run the Complete Workflow

The runner executes the stages sequentially and stops if a stage fails:

```
python run_workflow.py --list
python run_workflow.py
```

The first command only lists the stages; the second runs the complete analysis, regenerates report figures and tables, and runs the workflow checks. Execution overwrites notebook outputs, generated data, result files and saved models. Use a separate copy of the submission to preserve the supplied results. Each execution records Python and installed package versions in `results/runs/`. Leave-one-out evaluation and stacking involve many model fits and can take substantially longer than the other stages.

The execution order and main outputs are:

1. **Exploration (`eda`):** `notebooks/gallstone_eda.ipynb` explores the original data.
2. **Preparation (`prepare`):** `notebooks/transformation_feature_engineering.ipynb` exports training-fitted transformations and partitions to `data/processed/`.
3. **Holdout evaluation (`holdout`):** `notebooks/holdout_method.ipynb` writes metrics and predictions to `results/holdout/` and pipelines to `models/holdout/`.
4. **Leave-one-out evaluation (`lopo`):** `notebooks/leave_one_out_method.ipynb` evaluates 319 held-out rows per model family and writes `results/lopo/` and `models/lopo/`.

5. **Parameter tuning** (`tuning`): `notebooks/tuning_model.ipynb` writes search and evaluation results to `results/tuning/` and selected models to `models/tuned/`.

6. **Error analysis** (`errors`): `notebooks/error_analysis.ipynb` reads saved predictions and writes `results/error_analysis/`.

7. **Iteration comparison** (`patterns`): `scripts/iteration_pattern_analysis.py` compares the 20- and 10-predictor references and writes `results/iteration_patterns/`.

8. **Report outputs** (`figures`): `Report/build_modeling_figures.py` regenerates modelling figures and evidence tables in `Report/figures/modeling/` and `Report/generated/` from saved results.

9. **Verification** (`check`): unit checks cover preparation isolation, saved-pipeline prediction consistency and disease-positive label/false-negative-rate coding.

For interactive execution, run `python -m jupyter lab`, select the **Gallstone Iteration 3** kernel and use **Restart Kernel and Run All Cells** for each of the six notebooks in the order above. Complete each notebook before opening the next stage, then run:

```
python scripts/iteration_pattern_analysis.py
python Report/build_modeling_figures.py
python -m unittest discover -s tests -v
```

A.3 Resume Execution and Check Results

To restart at a failed stage, use `--from-stage`. To run just one stage, also specify `--to-stage`, for example:

```
python run_workflow.py --from-stage tuning
python run_workflow.py --from-stage holdout --to-stage holdout
python run_workflow.py --from-stage check
```

Stage selection does not rebuild prerequisites: tuning requires prepared data, and complete error analysis requires the preceding evaluation results. After changing data, seeds, preparation or model settings, rerun from `prepare` through the end; rerun exploration when raw data change. An interrupted stage is not a completed result.

Check `data/processed/manifest.json` and split-membership files for seeds 42, 7, 21, 84 and 123, with 223 training, 48 validation and 48 test rows per seed. A complete fixed holdout run produces 120 metric rows and 5,760 prediction rows; LOPO produces six metric rows and 1,914 predictions; tuning produces 25 metric rows and 1,200 predictions. These appear in the respective `results/holdout/`, `results/lopo/` and `results/tuning/` directories as `metrics.csv` and `predictions.csv`. The checks verify selected workflow invariants rather than every reported score.

Fixed seeds support repeatability within an environment, but package and platform differences can change exact results. Supplied pickle files require compatible library versions; if incompatible, regenerate preparation and models from `prepare` onward in the new environment. Evaluation must use the training-fitted partitions, not a new random split of the full-data `transformed_dataset.csv`.

A.4 Rebuild the Report PDF

With saved results present, regenerate the modelling figures and tables without retraining using the first command below. Install a LaTeX distribution with `latexmk`, `pdflatex`, Biber and the packages declared in `Report/main.tex`, then compile:

```
python Report/build_modeling_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -cd Report/main.tex
```

On PowerShell, enter the `latexmk` command on one line without the continuation backslash. The compiled report is `Report/main.pdf`. Retain all report source files, `references.bib`, `media/`, figures and generated tables. Figure generation does not update earlier exploratory images or manually written tables and discussion; reconcile those passages with any changed results before compiling the final report.